# Interpretive Kernel: From Occurrence Equivalence to Frame Equivalence

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**Abstract**

Many operational systems derive outputs by interpreting stable declarations against external frames that can change independently of those declarations. A recurring schedule, for example, may preserve the same stored rule while revisions to time-zone or calendar data alter its generated occurrences. Whether two frames are equivalent depends both on which interpretations are observed and on which output differences count as material. This paper formalizes that dependence. For finite frame and occurrence carriers, a finite probe family, and an interpretation function, the probe-indexed interpretation construction transports an occurrence equivalence relation to a frame equivalence relation by identifying two frame environments when every selected probe produces occurrence-equivalent outputs. The induced relation is the *interpretive kernel*: the partition of frame environments that the selected probes cannot distinguish. A reverse construction generates, from any declared frame regime, the least occurrence regime required to support it. The two maps form an adjunction between the lattices of occurrence and frame equivalence relations. Consequently, interpretive kernels preserve meets, generated demands preserve joins, and admissible occurrence regimes form a principal upper set with a least threshold. Pairwise thresholds yield an equivalence-relation-valued

generalized pseudoultrametric representation of the kernel family. Finite counterexamples show that unordered outputs require declared canonicalization, favorable matching need not preserve the required lattice structure, and frame-selection policies cannot be treated as identity regimes without equivalence-relation validation. A computational procedure constructs kernels, thresholds, and finite failure witnesses from a cached interpretation table using signatures, generating forests, and union-find. The framework is instantiated for recurring schedules using typed civil and resolved outputs, chronological canonicalization, finite observation horizons, and versioned frame environments. The result makes interpretive equivalence a declared, parameter-relative, and executable relation between rules of sameness on opposite sides of interpretation.

**Keywords:** structural explainability; interpretive kernels; occurrence equivalence; frame equivalence; observational equivalence; partition lattices; Galois connections; generalized ultrametrics; schedule semantics; time-zone data

# 1 Introduction

Many operational systems derive outputs by interpreting a stable declaration against an external frame. The declaration and the frame may change independently. A recurring schedule may retain the same rule, identifier, and stored representation while a revised time-zone or holiday-calendar artifact changes the occurrences it produces. Two deployments may therefore hold the same declaration and produce different outputs.

The converse also occurs. Distinct frame environments may produce the same observed occurrences for selected declarations and finite observation horizons. Whether those outputs count as equivalent may itself depend on whether the comparison preserves civil values, resolved instants, multiplicity, terminal status, or some coarser notion of occurrence. The question "do these frames produce the same output?" is therefore incomplete until the comparison declares both what is observed and which output differences count as material.

Scheduling is one instance of a broader class of interpreted systems.

Whenever a stable declaration is evaluated against an external, mutable, or versioned frame, the system contains at least two domains of sameness:

1. sameness among the interpretive frames; and
2. sameness among the outputs produced under those frames.

The two domains are related, but not interchangeable. A rule declaring two frame environments equivalent may require selected output differences to be ignored. A rule distinguishing two outputs may separate frames that another output regime would merge. This paper focuses on the relation between those choices.

*Neutral Substrates* (Case 2026a) establishes the neutrality-by-design constraint for systems operating under persistent interpretive disagreement.

Represented structure preserves stable reference without requiring agreement on every interpretation constructed over it.

*Referential Regimes* (Case 2026b) makes identity regime-relative and supplies the equivalence-relation framework needed to state different rules of sameness without collapsing them into one privileged answer.

*Operational Identity* (Case 2026c) represents declared and implemented rules of sameness as partitions of one finite carrier and compares them in the refinement lattice.

This work takes the next step. It studies how equivalence is transported across an interpretation function between two different finite carriers. One carrier contains frame environments; the other contains interpreted occurrence environments. A finite probe family selects which interpretations are observed. An occurrence regime declares which observed outputs count as equivalent. Together, these declarations induce an equivalence relation on the frame environments. That induced relation is the *interpretive kernel*.

## 1.1 Output Equivalence is Regime-Relative

Let

$$\mathcal{E}$$

be a finite carrier of frame environments, let

$$\mathcal{A}$$

be a finite carrier of canonical occurrence environments, and let

$$P$$

be a finite probe family.

An interpretation function

$$I : P \times \mathcal{E} \longrightarrow \mathcal{A}$$

assigns one occurrence environment to each probe and frame environment.

An occurrence regime

$$\rho \in \mathrm{Eq}(\mathcal{A})$$

determines which interpretation results count as equivalent.

The induced frame relation is

$$F \; K_P(\rho) \; F' \quad \Longleftrightarrow \quad \forall p \in P,\; I(p, F) \; \rho \; I(p, F').$$

The kernel is therefore relative to both the probe family and the occurrence regime. Changing either parameter may change the resulting frame partition.

Strict equality is one possible occurrence regime, but it is not the only one. A schedule comparison may distinguish civil values, resolved instants, multiplicity, terminal status, or other typed features. Another comparison may intentionally identify some of those differences. Neither choice is supplied by the interpretation function alone.

The reverse question is equally important. Suppose a system declares a frame regime

$$\varphi \in \mathrm{Eq}(\mathcal{E}).$$

Which occurrence distinctions must be ignored for every pair of frames identified by $\varphi$ to remain indistinguishable under the selected probes?

This paper constructs a generated occurrence regime

$$L_P(\varphi) \in \mathrm{Eq}(\mathcal{A})$$

and proves

$$L_P(\varphi) \sqsubseteq \rho \quad \Longleftrightarrow \quad \varphi \sqsubseteq K_P(\rho).$$

Thus

$$L_P \dashv K_P.$$

The left adjoint generates the least occurrence regime supporting a declared frame regime. The right adjoint returns the frame equivalence induced by a proposed occurrence regime. Their adjunction yields an exact threshold:

$$\rho \text{ supports } \varphi \quad \Longleftrightarrow \quad L_P(\varphi) \sqsubseteq \rho.$$

The threshold does not choose which frame or occurrence regime a system ought to adopt. It identifies the precise compatibility condition between the two declarations.

The same construction yields a pairwise interpretive demand

$$\delta_P(F, F'),$$

the least occurrence regime under which $F$ and $F'$ become interpretively equivalent.

These pairwise values form an $\mathrm{Eq}(\mathcal{A})$-valued generalized pseudoultrametric.

The representation is closely related to established correspondences between meet-preserving equivalence families and lattice-valued ultrametrics.

The contribution here is the finite interpretive specialization: the equivalence family is generated from probes, interpretation, and occurrence regimes, and each distance value has the operational meaning of a least supporting regime.

The construction also requires discipline about representation. An unordered family of occurrences cannot be compared index-wise without a declared canonicalization. Replacing canonicalization with optimization over alternative matchings does not repair the problem. Finite counterexamples show that such matching constructions may fail meet preservation and the strong triangle law. For schedule interpretation,

the paper therefore uses typed chronological canonicalization over finite observation horizons.

## 1.2 Contributions

This paper makes six contributions.

1. It defines a finite typed interpretation setting with separate frame and occurrence carriers, a finite probe family, and a regime-relative interpretive kernel.
2. It constructs the generated occurrence-regime map $L_P$ and proves the adjunction
$$L_P \dashv K_P.$$
The adjunction identifies the least occurrence regime supporting any declared frame regime and yields the corresponding meet- and join-preservation laws.
3. It defines pairwise interpretive demand and gives the generalized-pseudoultrametric representation of the kernel family, while locating that representation within the established lattice-valued-ultrametric correspondence.
4. It gives finite counterexamples showing why unordered presentations require declared canonicalization, why favorable matching does not preserve the required lattice structure, and why resolution policies cannot be admitted as frame regimes without equivalence-relation validation.
5. It provides a finite computational procedure for constructing partitions, kernels, thresholds, and pairwise demands using signatures, generating forests, and union-find, together with finite witnesses for failed admissibility and refinement claims.
6. It instantiates the framework for schedule interpretation using typed civil and resolved outputs, chronological canonicalization, finite recurrence horizons, and versioned frame environments, and states a preregistered evidence plan for testing nondegeneracy, regime sensitivity, threshold structure, and residuals.

The central result is therefore both algebraic and diagnostic. Equivalence declared on one side of an interpretation function induces equivalence on the other. The adjunction computes the exact boundary between the occurrence regimes that support a declared frame demand and those that do not. Because the carriers and probes are finite, that boundary is executable, and every failed admissibility claim

has a finite witness.

Section 2 states the imported interface from the prior Structural Explainability papers and the preconditions on a finite comparison artifact. Section 3 fixes the finite typed setting: occurrence and frame carriers, probes, the interpretation function, and the two regime lattices. Section 4 defines the interpretive kernel induced by a probe family and occurrence regime, and establishes its monotonicity and probe-union laws. Section 5 constructs the generated occurrence regime, proves the adjunction $L_P \dashv K_P$, and derives the meet- and join-preservation laws. Section 6 interprets the adjunction as an exact lattice-valued threshold between admissible and inadmissible occurrence regimes. Section 7 gives the pairwise interpretive demand and the generalized-pseudoultrametric representation. Section 8 gives finite counterexamples delimiting canonicalization, matching, and partition-hood requirements. Section 9 instantiates the construction for finite schedule interpretation over versioned frame artifacts. Section 10 gives the near-linear partition procedure and finite witness extraction. Section 11 states the preregistered evidence plan. Section 12 situates the work, Section 13 states its limits, and Section 14 concludes.

## 2 Imported Interface and Preconditions

This paper builds on three results from the Structural Explainability series.

*Neutral Substrates* separates the commitments required for stable shared reference from the causal, normative, and domain-specific interpretations that may be constructed over them. The substrate fixes what is jointly represented without selecting among the interpretations that may later be applied.

*Referential Regimes* establishes that preservation and invariance are relative to an explicitly declared basis. A transformation may preserve an object under one regime and distinguish it under another. The relevant question is therefore not whether two representations are invariant without qualification, but under which declared regime they remain equivalent.

*Operational Identity* represents declared and implemented rules of sameness as partitions of a common finite carrier and compares those partitions in the refinement lattice. That construction demonstrates that a system's effective equivalence relation can be made explicit and finitely compared without identifying equivalence with

equality of representation.

This work applies those commitments to interpretation. It studies two independently variable sources of equivalence: the regime under which occurrences are compared and the regime under which interpretive frames are compared. A finite probe family connects the two. The resulting construction determines which occurrence distinctions remain observable under a fixed family of probes and a fixed frame regime.

The earlier papers do not supply the occurrence regimes, frame regimes, probe families, interpretation functions, or interpretive kernels defined here. Those objects are introduced in Sections 3 and 4.

## 2.1 Imported Commitments

The present construction assumes the following commitments from the prior papers.

(I1) *Declared comparison.* Equivalence is evaluated relative to an explicitly declared regime.

(I2) *Separation of representation and interpretation.* A shared carrier may support multiple interpretations without incorporating any one interpretation into the carrier itself.

(I3) *Finite extensional comparison.* On the examined artifact, each regime induces a decidable equivalence relation and therefore a finite partition of its declared carrier.

(I4) *Typed interpretation.* The inputs, environmental parameters, and outputs of an interpretation are fixed by a declared signature.

(I5) *Declared output comparison.* Outputs are compared only through an explicitly declared canonicalization or comparison lifting.

## 2.2 Preconditions on the Comparison Artifact

The constructions of this paper are relative to a finite comparison artifact. The artifact supplies:

(P1) a finite occurrence carrier;

(P2) a finite frame-environment carrier;

(P3) decidable occurrence and frame regimes on those carriers;

(P4) a finite probe family;

(P5) a total interpretation function on the examined domain;

(P6) a declared output canonicalization or comparison lifting; and

(P7) any finite horizon required to generate the examined occurrences.

The finite horizon is part of the probe declaration. An unbounded recurrence specification may denote an indefinitely extensible family, but a finite probe evaluates only the occurrences generated within its declared horizon.

Failure to supply one of these components does not refute an interpretive equivalence. It leaves the corresponding kernel or threshold undefined for the examined artifact.

# 3 Mathematical Setting

This section fixes the finite typed setting used throughout the paper. The construction distinguishes occurrence environments, frame environments, and the probes under which one is interpreted relative to the other. It then represents occurrence and frame regimes as equivalence relations on their respective carriers.

## 3.1 Finite Carriers

**Definition 3.1** (Finite Comparison Artifact)**.** A *finite comparison artifact* consists of:

(a) a finite set $\mathcal{A}$ of occurrence environments;

(b) a finite set $\mathcal{E}$ of frame environments;

(c) a finite probe family $P$; and

(d) a total interpretation function

$$I : P \times \mathcal{E} \longrightarrow \mathcal{A}.$$

The finiteness requirement is relative to the examined artifact. It does not assert that the interpreted domain, the set of possible frame environments, or the family

of possible probes is globally finite. It asserts that the carriers and probes admitted to one comparison are finite and explicitly delimited.

For domains with indefinitely extensible behavior, finiteness is obtained through the probe declaration. A recurrence rule may admit occurrences beyond every fixed date, but a probe evaluates that rule only within its declared finite horizon. The resulting occurrence environment is therefore finite even when the underlying specification is unbounded.

*Remark* 3.2 (Artifact Relativity). Every partition, kernel, threshold, witness, and comparison in this paper is relative to a fixed finite comparison artifact.

Enlarging $\mathcal{A}$, enlarging $\mathcal{E}$, adding probes to $P$, extending a probe horizon, or revising the interpretation function produces a revised artifact on which the constructions must be recomputed.

## 3.2 Partitions and Refinement

Let $X$ be a finite set. Write $\mathrm{Eq}(X)$ for the set of equivalence relations on $X$.

Each $\alpha \in \mathrm{Eq}(X)$ induces the partition

$$X/\alpha = \big\{ [x]_\alpha \;\big|\; x \in X \big\}.$$

The relation and its induced partition will be used interchangeably when no ambiguity results.

**Definition 3.3** (Refinement Order)**.** For $\alpha, \beta \in \mathrm{Eq}(X)$, write

$$\alpha \sqsubseteq \beta$$

when

$$\alpha \subseteq \beta$$

as subsets of $X \times X$.

Equivalently, $\alpha \sqsubseteq \beta$ when every block of $X/\alpha$ is contained in a block of $X/\beta$. Thus $\alpha$ is finer than $\beta$, and $\beta$ is coarser than $\alpha$.

Under $\sqsubseteq$, $\mathrm{Eq}(X)$ is a finite lattice. Its least element is the diagonal relation

$$\bot_X = \{(x, x) \mid x \in X\},$$

and its greatest element is

$$\top_X = X \times X.$$

For $\alpha, \beta \in \mathrm{Eq}(X)$, their meet is intersection:

$$\alpha \wedge \beta = \alpha \cap \beta.$$

Their join is the equivalence closure of their union:

$$\alpha \vee \beta = \mathrm{EqCl}_X(\alpha \cup \beta).$$

More generally, for a relation $R \subseteq X \times X$, write

$$\mathrm{EqCl}_X(R)$$

for the least equivalence relation on $X$ containing $R$.

**Definition 3.4** (Generating Relation)**.** Let $\alpha \in \mathrm{Eq}(X)$. A relation $G \subseteq X \times X$ *generates* $\alpha$ when

$$\mathrm{EqCl}_X(G) = \alpha.$$

A generating relation need not contain every pair in $\alpha$. For a finite equivalence class, a connected set of generating pairs is sufficient. Section 10 uses this fact to replace all within-class pairs by a spanning forest.

## 3.3 Occurrence and Frame Environments

The two carriers represent different kinds of variation.

**Definition 3.5** (Occurrence Environment)**.** An *occurrence environment* $A \in \mathcal{A}$ is a finite canonical representation of the occurrences observed by a probe.

The representation contains every component declared relevant to occurrence comparison, including any ordering, multiplicity, boundary, or termination information required by the domain signature.

An occurrence environment is not necessarily a single occurrence. It may be a finite sequence, a finite indexed family, a bounded trace, or another finite structured output. Its internal form is fixed by the declared interpretation signature.

When occurrence environments contain ordered outputs, the order is part of their canonical representation. For the schedule interpretation of Section 9, that order is chronological. The comparison does not minimize over arbitrary bijections between generated occurrences.

A terminal or end tag is also part of the canonical representation when the domain distinguishes completion, truncation, or exhaustion of generated output. Differences in length or termination are then ordinary represented differences rather than exceptional cases outside the comparison.

**Definition 3.6** (Frame Environment)**.** A *frame environment* $F \in \mathcal{E}$ is a complete assignment to the frame variables of the declared interpretation signature.

Frame variables are parameters under which the same probe may be interpreted differently without changing the probe declaration itself.

A frame environment may include, for example, a resolution policy, a calendar convention, a time-zone database release, a locale, or another declared interpretive context. The abstract construction does not prescribe which variables belong to the frame. That choice is part of the domain interpretation.

**Definition 3.7** (Occurrence and Frame Regimes)**.** An *occurrence regime* is an equivalence relation

$$\rho \in \mathrm{Eq}(\mathcal{A}).$$

A *frame regime* is an equivalence relation

$$\varphi \in \mathrm{Eq}(\mathcal{E}).$$

An occurrence regime declares which occurrence environments count as equivalent. A frame regime declares which frame environments count as equivalent. Neither equivalence is inferred from raw representation equality unless equality is itself the declared regime.

The two regime lattices are

$$\mathrm{Eq}(\mathcal{A}) \qquad \text{and} \qquad \mathrm{Eq}(\mathcal{E}),$$

each ordered by refinement as in Definition 3.3.

## 3.4 Interpretation Functions

**Definition 3.8** (Interpretation Function)**.** For a finite probe family $P$, an *interpretation function* is a total function

$$I : P \times \mathcal{E} \longrightarrow \mathcal{A}.$$

For $p \in P$ and $F \in \mathcal{E}$, the value

$$I(p, F)$$

is the occurrence environment obtained by interpreting probe $p$ under frame environment $F$.

Totality is required on the examined artifact. A missing, invalid, or inapplicable interpretation result must therefore be represented by a declared value in $\mathcal{A}$ or excluded by a precondition on the artifact. It is not treated as an implicit absence of output.

The interpretation function may be factored through raw output. Let $\mathcal{O}$ be a finite raw-output carrier, let

$$J : P \times \mathcal{E} \longrightarrow \mathcal{O}$$

be a raw interpretation, and let

$$c : \mathcal{O} \longrightarrow \mathcal{A}$$

be a declared canonicalization. Then

$$I = c \circ J.$$

The abstract results depend on $I$, not on the internal procedure used to compute it.

For a fixed frame environment $F$, the section $I(\cdot, F) \colon P \to \mathcal{A}$ plays the role of a valuation function familiar from denotational semantics (Schmidt 1986, §4.1): it assigns to each probe its meaning as an occurrence environment. What distinguishes the present setting is that the frame is a parameter. The same probe is valued against many frames, and the object of study is the equivalence induced on frames

when two of them value every selected probe alike.

A domain realization must nevertheless declare how raw output is transformed into the occurrence environment on which its occurrence regimes are defined.

*Remark* 3.9 (Canonicalization Is Part of the Declaration). Canonicalization is not an optimization applied after comparison. It determines the carrier on which comparison is performed.

Two outputs that differ only by an undeclared presentation order may be mapped to the same canonical occurrence environment. Two outputs whose order is semantically declared remain distinguishable through that order.

Replacing declared canonicalization by minimization over all bijections defines a different comparison and may produce different kernels.

*Remark* 3.10 (Comparison Liftings). A domain may define an occurrence regime directly on $\mathcal{A}$, or obtain it by lifting an equivalence relation on simpler output values to sequences, finite families, or other structured occurrence environments.

Such a lifting must be declared before the interpretive kernel is computed. The kernel construction receives the resulting occurrence regime $\rho$; it does not choose the lifting or infer one from the observed outputs.

## 3.5 Probes

**Definition 3.11** (Probe)**.** A *probe* $p$ is a finite observation declaration sufficient to determine $I(p, F)$ for every frame environment $F \in \mathcal{E}$.

A probe fixes:

(a) the interpreted input;
(b) the finite observation horizon;
(c) the requested occurrence output; and
(d) any additional observation parameters not supplied by the frame environment.

The horizon is constitutive of the probe. Two declarations that differ only in horizon are distinct probes because they may produce different occurrence environments.

An unbounded recurrence specification does not produce an infinite occurrence environment under a probe. The probe horizon determines the finite generated

prefix, and the canonical occurrence environment records the declared boundary or end condition.

**Definition 3.12** (Probe Family)**.** A *probe family* $P$ is a finite set of probes evaluated against the same frame carrier $\mathcal{E}$ and occurrence carrier $\mathcal{A}$ through one interpretation function

$$I : P \times \mathcal{E} \longrightarrow \mathcal{A}.$$

The probe family is the observational basis of the comparison. A frame distinction invisible to one probe family may become visible when another probe is added. The interpretive kernels of Section 4 are therefore indexed by $P$.

*Remark* 3.13 (Both Parameters Are Required)*.* A probe family alone does not determine an interpretive kernel. The same probes may identify different frame environments under different occurrence regimes.

An occurrence regime alone does not determine an interpretive kernel. The same regime may identify different frame environments under different probe families.

Every kernel reported in this paper is therefore indexed by both a probe family $P$ and an occurrence regime $\rho$.

The standing mathematical data for the next section are thus

$$P, \qquad \mathcal{A}, \qquad \mathcal{E}, \qquad I : P \times \mathcal{E} \longrightarrow \mathcal{A},$$

together with an occurrence regime

$$\rho \in \mathrm{Eq}(\mathcal{A})$$

or a frame regime

$$\varphi \in \mathrm{Eq}(\mathcal{E}),$$

according to the direction of the construction.

# 4 Interpretive Kernels

The interpretation function assigns an occurrence environment to each probe and frame environment. An occurrence regime determines which of those outputs

count as equivalent. The resulting pattern of probe-relative equivalence induces an equivalence relation on the frame environments.

## 4.1 Kernel Induced by Probe Family and Occurrence Regime

Fix a finite comparison artifact

$$(P, \mathcal{A}, \mathcal{E}, I)$$

as in Definition 3.1, and let

$$\rho \in \mathrm{Eq}(\mathcal{A})$$

be an occurrence regime.

**Definition 4.1** (Probe Signature)**.** The *probe signature* induced by $P$ and $\rho$ is the function

$$\Sigma_{P,\rho} : \mathcal{E} \longrightarrow \big(\mathcal{A}/\rho\big)^P$$

defined by

$$\Sigma_{P,\rho}(F)(p) = [I(p, F)]_\rho$$

for every $F \in \mathcal{E}$ and $p \in P$.

The probe signature records, for each frame environment, the occurrence-equivalence class produced by every probe. Its codomain is a function space indexed by $P$, so the definition requires no arbitrary ordering of the probe family.

**Definition 4.2** (Interpretive Kernel)**.** The *interpretive kernel* induced by the probe family $P$ and occurrence regime $\rho$ is

$$K_P(\rho) = \ker(\Sigma_{P,\rho}).$$

Equivalently, for $F, F' \in \mathcal{E}$,

$$F \; K_P(\rho) \; F'$$

when

$$I(p, F) \; \rho \; I(p, F') \qquad \text{for every } p \in P.$$

Two frame environments lie in the same block of $K_P(\rho)$ when no probe in $P$ distinguishes them under the declared occurrence regime $\rho$.

**Proposition 4.3** (The Kernel Is a Frame Regime)**.** *For every finite probe family $P$ and occurrence regime $\rho \in \mathrm{Eq}(\mathcal{A})$, the relation*

$$K_P(\rho)$$

*is an equivalence relation on $\mathcal{E}$. Hence*

$$K_P(\rho) \in \mathrm{Eq}(\mathcal{E}).$$

*Proof.* The relation $K_P(\rho)$ is the kernel of the function $\Sigma_{P,\rho}$, so it is an equivalence relation.

Explicitly, reflexivity follows because $I(p, F)\ \rho\ I(p, F)$ for every $p \in P$.

Symmetry follows because $I(p, F)\ \rho\ I(p, F')$ implies $I(p, F')\ \rho\ I(p, F)$.

Transitivity follows because

$$I(p, F)\ \rho\ I(p, F') \qquad \text{and} \qquad I(p, F')\ \rho\ I(p, F'')$$

imply

$$I(p, F)\ \rho\ I(p, F'')$$

for every $p \in P$. □

**Definition 4.4** (Interpretive Partition)**.** The quotient

$$\pi_{P,\rho} = \mathcal{E}/K_P(\rho)$$

is the *interpretive partition* induced by $P$ and $\rho$.

Each block of $\pi_{P,\rho}$ contains the frame environments that produce the same probe signature up to the occurrence distinctions preserved by $\rho$.

The kernel reports observational equivalence relative to the declared comparison artifact. It does not assert that two frame environments are identical in representation, interchangeable for every purpose, or equivalent under probes outside $P$.

*Remark* 4.5 (Kernel Terminology)*.* The term *kernel* is literal. The relation $K_P(\rho)$ is the kernel equivalence of the quotient-valued signature map $\Sigma_{P,\rho}$.

It is also interpretation-relative. Replacing $I$, $P$, or $\rho$ may change the signature map and therefore its kernel.

## 4.2 Both Parameters Are Constitutive

The notation $K_P(\rho)$ retains both parameters because neither can be recovered from the other.

The probe family determines which interpretive differences are exercised. The occurrence regime determines which differences in the resulting occurrence environments count as observable.

**Proposition 4.6** (Probe Dependence)**.** *There exist a finite interpretation function, an occurrence regime $\rho$, and finite probe families*

$$P \subsetneq Q$$

*such that*

$$K_Q(\rho) \subsetneq K_P(\rho).$$

*Proof.* Let

$$\mathcal{E} = \{F_0, F_1\}, \qquad \mathcal{A} = \{a, b\},$$

and let $\rho = \bot_{\mathcal{A}}$ be equality on $\mathcal{A}$.

Take distinct probes $p$ and $q$, with

$$I(p, F_0) = I(p, F_1) = a,$$

and

$$I(q, F_0) = a, \qquad I(q, F_1) = b.$$

Let

$$P = \{p\}, \qquad Q = \{p, q\}.$$

Probe $p$ does not distinguish the two frame environments, so

$$K_P(\rho) = \top_{\mathcal{E}}.$$

Probe $q$ distinguishes them under equality, so

$$K_Q(\rho) = \bot_{\mathcal{E}}.$$

Therefore

$$K_Q(\rho) \subsetneq K_P(\rho).$$

□

**Proposition 4.7** (Occurrence-Regime Dependence)**.** *There exist a finite probe family $P$ and occurrence regimes*

$$\rho \subsetneq \sigma$$

*such that*

$$K_P(\rho) \subsetneq K_P(\sigma).$$

*Proof.* Let

$$\mathcal{E} = \{F_0, F_1\}, \qquad \mathcal{A} = \{a, b\},$$

and let $P = \{p\}$.

Define

$$I(p, F_0) = a, \qquad I(p, F_1) = b.$$

Let

$$\rho = \bot_{\mathcal{A}}$$

and

$$\sigma = \top_{\mathcal{A}}.$$

Under $\rho$, the outputs $a$ and $b$ are distinct, so

$$K_P(\rho) = \bot_{\mathcal{E}}.$$

Under $\sigma$, every occurrence environment is equivalent, so

$$K_P(\sigma) = \top_{\mathcal{E}}.$$

Hence

$$K_P(\rho) \subsetneq K_P(\sigma).$$

□

Propositions 4.6 and 4.7 show that neither parameter is decorative.

Reporting only $P$ omits the rule under which outputs were compared. Reporting only $\rho$ omits the observations under which frame equivalence was tested. A reported interpretive kernel must identify both.

*Remark* 4.8 (No Unqualified Frame Equivalence). The construction does not define an unqualified statement that two frame environments are interpretively equivalent.

It defines the indexed statement

$$F \; K_P(\rho) \; F',$$

meaning that the frame environments agree under the probes in $P$ up to the occurrence distinctions retained by $\rho$.

## 4.3 Canonicalization and Comparison Liftings

The kernel compares elements of the occurrence carrier $\mathcal{A}$. The construction therefore begins only after raw interpretive output has been converted into the canonical occurrence representation on which the occurrence regime is defined.

Suppose the domain realization supplies a raw-output carrier $\mathcal{O}$, a raw interpretation

$$J : P \times \mathcal{E} \longrightarrow \mathcal{O},$$

and a canonicalization

$$c : \mathcal{O} \longrightarrow \mathcal{A}.$$

Then

$$I = c \circ J,$$

and Definition 4.2 becomes

$$F \; K_P(\rho) \; F' \quad \Longleftrightarrow \quad c(J(p, F)) \; \rho \; c(J(p, F')) \text{ for every } p \in P.$$

Canonicalization fixes the represented structure before comparison begins. It may normalize distinctions that the domain declares representational, but it must preserve distinctions that the domain declares interpretively relevant.

**Definition 4.9** (Comparison Lifting)**.** Let $V$ be a finite component-value carrier and let

$$\delta \in \mathrm{Eq}(V)$$

be a component equivalence.

A *comparison lifting* for the occurrence carrier $\mathcal{A}$ assigns to $\delta$ an occurrence regime

$$\widehat{\delta} \in \mathrm{Eq}(\mathcal{A}).$$

The lifting states how component equivalence extends to complete occurrence environments. For sequence-valued environments, the lifting may require equal length, index-wise component equivalence, and agreement of declared terminal information. For other structured outputs, a different lifting may be appropriate.

The kernel receives the lifted occurrence regime

$$\widehat{\delta}$$

and computes

$$K_P(\widehat{\delta}).$$

The component relation $\delta$ alone does not determine the comparison until its lifting to $\mathcal{A}$ has been declared.

**Definition 4.10** (Monotone Comparison Lifting)**.** A comparison lifting is *monotone* when

$$\delta \sqsubseteq \delta'$$

implies

$$\widehat{\delta} \sqsubseteq \widehat{\delta'}.$$

Monotonicity says that weakening component comparison cannot make the lifted comparison stricter. Any stronger structural laws required of a domain lifting must be stated as part of that domain's declared comparison protocol.

*Remark* 4.11 (Canonical Order Is Not Matching)*.* When the occurrence environment carries a declared canonical order, comparison is performed in that order.

The construction does not minimize over bijections between the members of two occurrence environments. Such minimization would quotient away the declared order and define a different occurrence regime.

In the schedule interpretation, generated occurrences are placed in chronological order. Comparison is then index-wise under the declared lifting, with length and terminal information retained as components of the canonical occurrence environment.

*Remark* 4.12 (Canonicalization Precedes the Kernel). Canonicalization and comparison lifting are not operations performed by $K_P$.

They determine the occurrence carrier and the occurrence regime supplied to $K_P$. The kernel then transports that declared output comparison to a frame regime through the interpretation function.

## 4.4 Monotonicity

The interpretive-kernel construction is monotone in the occurrence regime.

**Proposition 4.13** (Monotonicity in the Occurrence Regime)**.** *Let*

$$\rho, \sigma \in \mathrm{Eq}(\mathcal{A}).$$

*If*

$$\rho \sqsubseteq \sigma,$$

*then*

$$K_P(\rho) \sqsubseteq K_P(\sigma).$$

*Proof.* Suppose

$$F \; K_P(\rho) \; F'.$$

Then

$$I(p, F) \; \rho \; I(p, F')$$

for every $p \in P$.

Because

$$\rho \subseteq \sigma,$$

it follows that

$$I(p, F)\ \sigma\ I(p, F')$$

for every $p \in P$.

Hence

$$F\ K_P(\sigma)\ F',$$

so

$$K_P(\rho) \subseteq K_P(\sigma).$$

□

A coarser occurrence regime identifies at least as many occurrence environments. The induced interpretive kernel therefore identifies at least as many frame environments.

The dependence on the probe family has the opposite direction under ordinary set inclusion.

**Proposition 4.14** (Probe-Family Extension)**.** *Let $P$ and $Q$ be finite probe families for the same interpretation function, with*

$$P \subseteq Q.$$

*Then for every occurrence regime $\rho$,*

$$K_Q(\rho) \sqsubseteq K_P(\rho).$$

*Proof.* Suppose

$$F\ K_Q(\rho)\ F'.$$

Then

$$I(q, F)\ \rho\ I(q, F')$$

for every $q \in Q$.

Because $P \subseteq Q$, the same condition holds for every $p \in P$. Therefore

$$F\ K_P(\rho)\ F'.$$

□

Adding probes can preserve a frame equivalence or split an existing equivalence class. It cannot merge frame environments that the smaller probe family distinguished.

**Proposition 4.15** (Probe-Union Meet Law)**.** *For finite probe families $P$ and $Q$ and every occurrence regime $\rho$,*

$$K_{P\cup Q}(\rho) = K_P(\rho) \wedge K_Q(\rho).$$

*Proof.* For $F, F' \in \mathcal{E}$,

$$F \; K_{P\cup Q}(\rho) \; F'$$

holds when

$$I(p, F) \; \rho \; I(p, F')$$

for every $p \in P \cup Q$.

This is equivalent to the conjunction of

$$F \; K_P(\rho) \; F'$$

and

$$F \; K_Q(\rho) \; F'.$$

Therefore

$$K_{P\cup Q}(\rho) = K_P(\rho) \cap K_Q(\rho).$$

Meet in $\mathrm{Eq}(\mathcal{E})$ is intersection, so the result follows. □

**Corollary 4.16** (Kernel as a Meet of Single-Probe Kernels)**.** *For every finite probe family $P$ and occurrence regime $\rho$,*

$$K_P(\rho) = \bigwedge_{p\in P} K_{\{p\}}(\rho).$$

**Corollary 4.17** (Boundary Kernels)**.** *For every finite probe family $P$,*

$$K_P(\top_{\mathcal{A}}) = \top_{\mathcal{E}}.$$

*For the empty probe family,*

$$K_{\varnothing}(\rho) = \top_{\mathcal{E}}$$

*for every occurrence regime $\rho$.*

The first equality holds because the indiscrete occurrence regime identifies every pair of occurrence environments. The second holds because the universal condition over an empty probe family is vacuously satisfied.

For equality on the occurrence carrier,

$$K_P(\bot_{\mathcal{A}})$$

identifies those frame environments that produce equal canonical occurrence environments for every probe in $P$.

The map

$$K_P : \mathrm{Eq}(\mathcal{A}) \longrightarrow \mathrm{Eq}(\mathcal{E})$$

is therefore a monotone map between finite partition lattices. The next section constructs the map in the opposite direction from a declared frame regime and establishes the relation between the two constructions.

# 5 The Adjoint Construction

The interpretive-kernel construction sends an occurrence regime to the frame regime that it induces under a fixed probe family. This section constructs the map in the opposite direction.

A declared frame regime requires certain pairs of frame environments to remain indistinguishable. For each probe, that requirement determines pairs of occurrence environments that must be treated as equivalent. Taking the equivalence closure of those required pairs yields the least occurrence regime capable of supporting the declared frame regime.

## 5.1 Generated Occurrence Regime

Fix a finite comparison artifact

$$(P, \mathcal{A}, \mathcal{E}, I)$$

and let

$$\varphi \in \mathrm{Eq}(\mathcal{E})$$

be a frame regime.

**Definition 5.1** (Frame-Generated Output Relation)**.** The *frame-generated output relation* induced by $P$ and $\varphi$ is

$$R_{P,\varphi} = \big\{ \big(I(p,F), I(p,F')\big) \;\big|\; p \in P \text{ and } F \;\varphi\; F' \big\}.$$

The relation $R_{P,\varphi}$ contains every pair of occurrence environments that the interpretation function produces from frame environments the declared frame regime requires to remain equivalent.

The relation need not itself be an equivalence relation. Different frame pairs may produce overlapping output pairs, and the required output equivalence may arise only after symmetry and transitivity are imposed.

**Definition 5.2** (Generated Occurrence Regime)**.** The *generated occurrence regime* induced by the probe family $P$ and frame regime $\varphi$ is

$$L_P(\varphi) = \mathrm{EqCl}_{\mathcal{A}}\big(R_{P,\varphi}\big).$$

By construction,

$$L_P(\varphi) \in \mathrm{Eq}(\mathcal{A}).$$

The map

$$L_P : \mathrm{Eq}(\mathcal{E}) \longrightarrow \mathrm{Eq}(\mathcal{A})$$

therefore sends a frame regime to an occurrence regime.

**Proposition 5.3** (Least Supporting Occurrence Regime)**.** *Let*

$$\varphi \in \mathrm{Eq}(\mathcal{E}).$$

*Then $L_P(\varphi)$ has the following properties:*

*(a) for every $p \in P$ and $F, F' \in \mathcal{E}$,*

$$F \;\varphi\; F' \quad\Longrightarrow\quad I(p,F)\; L_P(\varphi)\; I(p,F');$$

*(b) if*

$$\rho \in \mathrm{Eq}(\mathcal{A})$$

*also satisfies*

$$F \ \varphi \ F' \quad \Longrightarrow \quad I(p,F) \ \rho \ I(p,F')$$

*for every $p \in P$, then*

$$L_P(\varphi) \sqsubseteq \rho.$$

*Proof.* Every pair

$$\big(I(p,F), I(p,F')\big)$$

with $F \ \varphi \ F'$ lies in $R_{P,\varphi}$, and therefore lies in its equivalence closure $L_P(\varphi)$. This proves the first statement.

For the second, suppose $\rho$ contains every such pair. Then

$$R_{P,\varphi} \subseteq \rho.$$

Because $\rho$ is an equivalence relation, it contains the least equivalence relation generated by $R_{P,\varphi}$. Hence

$$L_P(\varphi) = \mathrm{EqCl}_{\mathcal{A}}\big(R_{P,\varphi}\big) \subseteq \rho.$$

□

Thus $L_P(\varphi)$ does not introduce an arbitrary occurrence comparison. It contains the equivalences forced by the frame regime under the probes in $P$, together with those required by equivalence closure.

**Proposition 5.4** (Monotonicity of the Generated Regime)**.** *Let*

$$\varphi, \psi \in \mathrm{Eq}(\mathcal{E}).$$

*If*

$$\varphi \sqsubseteq \psi,$$

*then*

$$L_P(\varphi) \sqsubseteq L_P(\psi).$$

*Proof.* If

$$\varphi \subseteq \psi,$$

then every pair used to construct $R_{P,\varphi}$ is also used to construct $R_{P,\psi}$. Therefore

$$R_{P,\varphi} \subseteq R_{P,\psi}.$$

Equivalence closure is monotone, so

$$L_P(\varphi) \subseteq L_P(\psi).$$

□

## 5.2 Generating-Set Reduction

Definition 5.1 ranges over every pair in the frame equivalence relation. That full pair set is not required.

A generating relation for the frame regime is sufficient to generate the same occurrence regime.

**Definition 5.5** (Probe Image of a Frame Relation)**.** For a relation

$$G \subseteq \mathcal{E} \times \mathcal{E},$$

define

$$R_{P,G} = \big\{ \big(I(p,F), I(p,F')\big) \;\big|\; p \in P \text{ and } (F,F') \in G \big\}.$$

**Theorem 5.6** (Generating-Set Reduction)**.** *Let*

$$\varphi \in \mathrm{Eq}(\mathcal{E}),$$

*and let*

$$G \subseteq \mathcal{E} \times \mathcal{E}$$

*generate $\varphi$, so that*

$$\mathrm{EqCl}_{\mathcal{E}}(G) = \varphi.$$

*Then*

$$\mathrm{EqCl}_{\mathcal{A}}\big(R_{P,G}\big) = L_P(\varphi).$$

*Proof.* Because

$$G \subseteq \varphi,$$

Definition 5.5 gives

$$R_{P,G} \subseteq R_{P,\varphi}.$$

Therefore

$$\mathrm{EqCl}_{\mathcal{A}}\big(R_{P,G}\big) \subseteq L_P(\varphi).$$

For the reverse inclusion, take a generating pair

$$\big(I(p,F), I(p,F')\big) \in R_{P,\varphi}.$$

Then

$$F \;\varphi\; F'.$$

Since

$$\mathrm{EqCl}_{\mathcal{E}}(G) = \varphi,$$

there exists a finite sequence

$$F = F_0, F_1, \ldots, F_k = F'$$

such that each consecutive pair is contained in $G$ in one direction or the other.

Fixing the same probe $p$, each consecutive output pair

$$\big(I(p,F_i), I(p,F_{i+1})\big)$$

therefore lies in $R_{P,G}$ in one direction or the other.

Symmetry and transitivity of

$$\mathrm{EqCl}_{\mathcal{A}}\big(R_{P,G}\big)$$

give

$$I(p,F)\ \mathrm{EqCl}_{\mathcal{A}}\left(R_{P,G}\right)\ I(p,F').$$

Every generator in $R_{P,\varphi}$ therefore lies in

$$\mathrm{EqCl}_{\mathcal{A}}\big(R_{P,G}\big).$$

Taking equivalence closures yields

$$L_P(\varphi) \subseteq \mathrm{EqCl}_{\mathcal{A}}\big(R_{P,G}\big).$$

□

**Corollary 5.7** (Spanning-Forest Reduction)**.** *Let*

$$\varphi \in \mathrm{Eq}(\mathcal{E}).$$

*For each block of*

$$\mathcal{E}/\varphi,$$

*choose a spanning tree on the members of that block, and let $G_\varphi$ be the union of the selected edge sets.*

*Then*

$$\mathrm{EqCl}_{\mathcal{E}}\big(G_\varphi\big) = \varphi,$$

*and therefore*

$$L_P(\varphi) = \mathrm{EqCl}_{\mathcal{A}}\big(R_{P,G_\varphi}\big).$$

*The number of selected frame pairs is*

$$|\mathcal{E}| - |\mathcal{E}/\varphi|.$$

*Proof.* A spanning tree connects every pair of vertices in its selected equivalence class, while no selected edge connects different classes. The equivalence closure of the forest is therefore $\varphi$.

A block containing $n$ frame environments contributes $n-1$ edges. Summing over the blocks gives

$$\sum_{C \in \mathcal{E}/\varphi} \big(|C| - 1\big) = |\mathcal{E}| - |\mathcal{E}/\varphi|.$$

The result then follows from Theorem 5.6. □

The reduction replaces a possible quadratic enumeration of all pairs within frame-equivalence classes by a linear-size generating relation. For each selected frame pair, the interpretation need be compared only under the probes in $P$.

## 5.3 The Adjunction

The maps

$$L_P : \mathrm{Eq}(\mathcal{E}) \longrightarrow \mathrm{Eq}(\mathcal{A})$$

and

$$K_P : \mathrm{Eq}(\mathcal{A}) \longrightarrow \mathrm{Eq}(\mathcal{E})$$

form an adjoint pair.

**Theorem 5.8** (Interpretive Adjunction)**.** *For every frame regime*

$$\varphi \in \mathrm{Eq}(\mathcal{E})$$

*and occurrence regime*

$$\rho \in \mathrm{Eq}(\mathcal{A}),$$

$$L_P(\varphi) \sqsubseteq \rho \quad \Longleftrightarrow \quad \varphi \sqsubseteq K_P(\rho).$$

*Hence*

$$L_P \dashv K_P.$$

*Proof.* Suppose first that

$$L_P(\varphi) \sqsubseteq \rho.$$

Let

$$F \; \varphi \; F'.$$

By Proposition 5.3,

$$I(p, F) \; L_P(\varphi) \; I(p, F')$$

for every $p \in P$.

Since

$$L_P(\varphi) \subseteq \rho,$$

it follows that

$$I(p, F) \; \rho \; I(p, F')$$

for every $p \in P$.

Therefore

$$F \; K_P(\rho) \; F',$$

and hence

$$\varphi \sqsubseteq K_P(\rho).$$

Conversely, suppose

$$\varphi \sqsubseteq K_P(\rho).$$

If

$$F \; \varphi \; F',$$

then

$$F \; K_P(\rho) \; F'.$$

By Definition 4.2,

$$I(p, F) \; \rho \; I(p, F')$$

for every $p \in P$.

Thus

$$R_{P,\varphi} \subseteq \rho.$$

Since $\rho$ is an equivalence relation, it contains the equivalence closure of $R_{P,\varphi}$. Therefore

$$L_P(\varphi) \sqsubseteq \rho.$$

□

The theorem states the same compatibility condition in two equivalent ways.

The left-hand statement asks whether the occurrence regime $\rho$ contains every output equivalence required by the frame regime $\varphi$.

The right-hand statement asks whether the interpretive kernel induced by $\rho$ contains every frame equivalence declared by $\varphi$.

**Corollary 5.9** (Unit and Counit)**.** *For every frame regime*

$$\varphi \in \mathrm{Eq}(\mathcal{E}),$$

$$\varphi \sqsubseteq K_P\big(L_P(\varphi)\big).$$

*For every occurrence regime*

$$\rho \in \mathrm{Eq}(\mathcal{A}),$$

$$L_P\big(K_P(\rho)\big) \sqsubseteq \rho.$$

*Proof.* Apply Theorem 5.8 first with

$$\rho = L_P(\varphi),$$

using reflexivity of the refinement order, and then with

$$\varphi = K_P(\rho).$$

□

The first inequality says that the least occurrence regime generated by a frame regime supports at least that frame regime when transported back through the kernel.

The second says that regenerating an occurrence regime from its induced frame kernel introduces no occurrence equivalence that was absent from the original regime.

*Remark* 5.10 (Equality Need Not Hold). Neither inequality in Corollary 5.9 is asserted to be an equality.

The probe family may be unable to observe every distinction carried by a frame regime, so

$$\varphi \subsetneq K_P\big(L_P(\varphi)\big)$$

may hold.

An occurrence regime may also contain distinctions unrelated to the outputs generated by $P$, so

$$L_P\big(K_P(\rho)\big) \subsetneq \rho$$

may hold.

## 5.4 Preservation Laws

As a right adjoint, the interpretive-kernel map preserves meets. As a left adjoint, the generated-regime map preserves joins.

**Theorem 5.11** (Meet Preservation)**.** *Let*

$$\{\rho_i\}_{i\in J} \subseteq \mathrm{Eq}(\mathcal{A})$$

*be a family of occurrence regimes. Then*

$$K_P\left(\bigwedge_{i\in J} \rho_i\right) = \bigwedge_{i\in J} K_P(\rho_i).$$

*Proof.* For $F, F' \in \mathcal{E}$,

$$F \; K_P\left(\bigwedge_{i\in J} \rho_i\right) F'$$

holds when

$$I(p,F) \bigwedge_{i\in J} \rho_i \; I(p,F')$$

for every $p \in P$.

Since meet in $\mathrm{Eq}(\mathcal{A})$ is intersection, this holds when

$$I(p,F) \; \rho_i \; I(p,F')$$

for every $p \in P$ and every $i \in J$.

That condition is equivalent to

$$F \; K_P(\rho_i) \; F'$$

for every $i \in J$, which is

$$F \bigwedge_{i\in J} K_P(\rho_i) \; F'.$$

□

**Theorem 5.12** (Join Preservation)**.** *Let*

$$\{\varphi_i\}_{i\in J} \subseteq \mathrm{Eq}(\mathcal{E})$$

*be a family of frame regimes. Then*

$$L_P\left(\bigvee_{i\in J} \varphi_i\right) = \bigvee_{i\in J} L_P(\varphi_i).$$

*Proof.* Let

$$\rho \in \mathrm{Eq}(\mathcal{A}).$$

By Theorem 5.8,

$$L_P\left(\bigvee_{i\in J} \varphi_i\right) \sqsubseteq \rho$$

holds when

$$\bigvee_{i\in J} \varphi_i \sqsubseteq K_P(\rho).$$

By the defining property of join, this holds when

$$\varphi_i \sqsubseteq K_P(\rho)$$

for every $i \in J$.

Applying the adjunction to each $i$, this is equivalent to

$$L_P(\varphi_i) \sqsubseteq \rho$$

for every $i \in J$.

By the defining property of join, the latter holds when

$$\bigvee_{i\in J} L_P(\varphi_i) \sqsubseteq \rho.$$

The two left-hand elements therefore have the same upper bounds, and hence are equal. □

**Corollary 5.13** (Boundary Preservation)**.**

$$K_P\big(\top_{\mathcal{A}}\big) = \top_{\mathcal{E}},$$

*and*

$$L_P\big(\bot_{\mathcal{E}}\big) = \bot_{\mathcal{A}}.$$

The preservation laws are directional. The adjunction guarantees meet preservation by $K_P$ and join preservation by $L_P$. It does not by itself establish join preservation by $K_P$ or meet preservation by $L_P$.

## 5.5 Probe-Family Duality

Changing the probe family acts dually on the two constructions.

Adding probes makes the interpretive kernel finer, because more observations must agree. The same addition makes the generated occurrence regime coarser, because more output-equivalence requirements are imposed.

**Theorem 5.14** (Probe-Union Duality)**.** *Let $P$ and $Q$ be finite probe families for the same occurrence carrier, frame carrier, and interpretation function.*

*For every occurrence regime*

$$\rho \in \mathrm{Eq}(\mathcal{A}),$$

$$K_{P\cup Q}(\rho) = K_P(\rho) \wedge K_Q(\rho).$$

*For every frame regime*

$$\varphi \in \mathrm{Eq}(\mathcal{E}),$$

$$L_{P\cup Q}(\varphi) = L_P(\varphi) \vee L_Q(\varphi).$$

*Proof.* The first equality is Proposition 4.15.

For the second, Definition 5.1 gives

$$R_{P\cup Q,\varphi} = R_{P,\varphi} \cup R_{Q,\varphi}.$$

Therefore

$$L_{P\cup Q}(\varphi) = \mathrm{EqCl}_{\mathcal{A}}\big(R_{P,\varphi} \cup R_{Q,\varphi}\big).$$

Join in $\mathrm{Eq}(\mathcal{A})$ is the equivalence closure of union, so

$$L_{P\cup Q}(\varphi) = L_P(\varphi) \vee L_Q(\varphi).$$

□

**Corollary 5.15** (Finite Probe-Family Decomposition)**.** *Let*

$$\{P_i\}_{i\in J}$$

*be a finite family of finite probe families.*

*For every occurrence regime $\rho$,*

$$K_{\bigcup_{i\in J} P_i}(\rho) = \bigwedge_{i\in J} K_{P_i}(\rho).$$

*For every frame regime $\varphi$,*

$$L_{\bigcup_{i\in J} P_i}(\varphi) = \bigvee_{i\in J} L_{P_i}(\varphi).$$

**Corollary 5.16** (Empty Probe Family)**.** *For every occurrence regime $\rho$,*

$$K_{\varnothing}(\rho) = \top_{\mathcal{E}}.$$

*For every frame regime $\varphi$,*

$$L_{\varnothing}(\varphi) = \bot_{\mathcal{A}}.$$

The empty probe family distinguishes no frame environments and imposes no nontrivial output-equivalence requirements.

The adjunction and its preservation laws provide the algebraic basis for the threshold construction. A declared frame regime $\varphi$ is supported by an occurrence regime $\rho$ when

$$L_P(\varphi) \sqsubseteq \rho,$$

equivalently when

$$\varphi \sqsubseteq K_P(\rho).$$

The next section studies the boundary between occurrence regimes that satisfy this condition and those that do not.

# 6 Thresholds

The adjunction characterizes when an occurrence regime is sufficiently coarse to support a declared frame regime.

That characterization has the form of a threshold. For each probe family and frame regime, there is a least occurrence regime that identifies every output pair whose equivalence is required by the frame regime. Every coarser occurrence regime is also

sufficient, while every regime not lying above that least element is insufficient.

The threshold is therefore lattice-valued. It is an equivalence relation on the occurrence carrier, not a scalar score or a position in a total order.

## 6.1 Admissible Occurrence Regimes

Fix a finite comparison artifact, a probe family $P$, and a frame regime

$$\varphi \in \mathrm{Eq}(\mathcal{E}).$$

**Definition 6.1** (Admissible Occurrence Regime)**.** An occurrence regime

$$\rho \in \mathrm{Eq}(\mathcal{A})$$

is *admissible for* $\varphi$ *under* $P$ when

$$\varphi \sqsubseteq K_P(\rho).$$

Write

$$\mathrm{Adm}_P(\varphi) = \{\rho \in \mathrm{Eq}(\mathcal{A}) \mid \varphi \sqsubseteq K_P(\rho)\}$$

for the set of admissible occurrence regimes.

Admissibility means that every pair of frame environments identified by $\varphi$ also lies in the interpretive kernel induced by $\rho$. Equivalently, for every $F, F' \in \mathcal{E}$,

$$F \ \varphi \ F'$$

implies

$$I(p, F) \ \rho \ I(p, F')$$

for every $p \in P$.

**Proposition 6.2** (Adjoint Characterization of Admissibility)**.** *For every occurrence regime*

$$\rho \in \mathrm{Eq}(\mathcal{A}),$$

$$\rho \in \mathrm{Adm}_P(\varphi)$$

*if and only if*

$$L_P(\varphi) \sqsubseteq \rho.$$

*Proof.* By Definition 6.1, $\rho$ is admissible when

$$\varphi \sqsubseteq K_P(\rho).$$

By Theorem 5.8, this holds when

$$L_P(\varphi) \sqsubseteq \rho.$$

□

Thus

$$\mathrm{Adm}_P(\varphi) = \{\rho \in \mathrm{Eq}(\mathcal{A}) \mid L_P(\varphi) \sqsubseteq \rho\}.$$

The admissible occurrence regimes form the principal upper set generated by $L_P(\varphi)$. Admissibility is therefore not an arbitrary predicate over the occurrence-regime lattice.

**Proposition 6.3** (Upward Closure of Admissibility)**.** *Let*

$$\rho, \sigma \in \mathrm{Eq}(\mathcal{A}).$$

*If*

$$\rho \in \mathrm{Adm}_P(\varphi) \qquad \textit{and} \qquad \rho \sqsubseteq \sigma,$$

*then*

$$\sigma \in \mathrm{Adm}_P(\varphi).$$

*Proof.* Proposition 6.2 gives

$$L_P(\varphi) \sqsubseteq \rho.$$

Combining this with

$$\rho \sqsubseteq \sigma$$

and transitivity yields

$$L_P(\varphi) \sqsubseteq \sigma.$$

The same proposition then gives

$$\sigma \in \mathrm{Adm}_P(\varphi).$$

□

Coarsening an admissible occurrence regime cannot invalidate the frame equivalences it already supports. It may identify additional occurrence environments and thereby induce additional frame equivalences, but it cannot remove an equivalence required by $\varphi$.

## 6.2 Lattice-Valued Thresholds

**Definition 6.4** (Interpretive Threshold)**.** The *interpretive threshold* of a frame regime $\varphi$ under probe family $P$ is

$$\Theta_P(\varphi) = L_P(\varphi).$$

The threshold is an element of

$$\mathrm{Eq}(\mathcal{A}).$$

It is therefore a partition of the occurrence carrier rather than a numerical value.

**Theorem 6.5** (Exact Threshold Characterization)**.** *For every frame regime $\varphi$ and occurrence regime $\rho$,*

$$\rho \in \mathrm{Adm}_P(\varphi) \quad \Longleftrightarrow \quad \Theta_P(\varphi) \sqsubseteq \rho.$$

*Proof.* Substitute

$$\Theta_P(\varphi) = L_P(\varphi)$$

into Proposition 6.2. □

The theorem gives the threshold its exact meaning.

An occurrence regime is sufficient precisely when it contains every occurrence equivalence generated by the frame regime under the probes in $P$.

A regime that is not above the threshold is not admissible. It either omits a required threshold pair or is incomparable with the threshold and therefore still omits at least one required equivalence.

**Corollary 6.6** (Threshold Is Admissible)**.** *For every frame regime $\varphi$,*

$$\Theta_P(\varphi) \in \mathrm{Adm}_P(\varphi).$$

*Proof.* Reflexivity gives

$$\Theta_P(\varphi) \sqsubseteq \Theta_P(\varphi).$$

Apply Theorem 6.5. □

**Corollary 6.7** (Threshold Is the Finest Admissible Regime)**.** *For every*

$$\rho \in \mathrm{Adm}_P(\varphi),$$

$$\Theta_P(\varphi) \sqsubseteq \rho.$$

*Hence $\Theta_P(\varphi)$ is the least element of $\mathrm{Adm}_P(\varphi)$ under the refinement order and the finest admissible occurrence regime.*

The threshold introduces no occurrence equivalence beyond those forced by the declared frame regime and the selected probes. Any strictly coarser admissible regime contains additional occurrence equivalences not required by that demand.

**Proposition 6.8** (Monotonicity of Thresholds)**.** *Let*

$$\varphi, \psi \in \mathrm{Eq}(\mathcal{E}).$$

*If*

$$\varphi \sqsubseteq \psi,$$

*then*

$$\Theta_P(\varphi) \sqsubseteq \Theta_P(\psi).$$

*Proof.* By Definition 6.4,

$$\Theta_P(\varphi) = L_P(\varphi)$$

and

$$\Theta_P(\psi) = L_P(\psi).$$

The result follows from Proposition 5.4. □

A coarser frame regime identifies more frame pairs and therefore makes a stronger invariance demand. Its threshold contains at least the occurrence equivalences required by every finer frame regime below it.

**Proposition 6.9** (Probe Extension Raises the Threshold)**.** *Let*

$$P \subseteq Q$$

*be finite probe families. Then for every frame regime $\varphi$,*

$$\Theta_P(\varphi) \sqsubseteq \Theta_Q(\varphi).$$

*Proof.* Since

$$Q = P \cup Q,$$

Theorem 5.14 gives

$$\Theta_Q(\varphi) = \Theta_P(\varphi) \vee \Theta_Q(\varphi).$$

Hence

$$\Theta_P(\varphi) \sqsubseteq \Theta_Q(\varphi).$$

□

Adding probes may reveal additional occurrence equivalences that are required to preserve the same frame regime. It can leave the threshold unchanged or move it upward to a coarser partition. It cannot make the threshold finer.

*Remark* 6.10 (Why the Threshold Is Not Scalar). The occurrence-regime lattice is generally only partially ordered.

Two thresholds may require different occurrence equivalences without either containing the other. Assigning them scalar levels would impose a linear comparison not supplied by the mathematics.

Threshold reporting must therefore retain the partition itself, or an equivalent finite representation of it, together with the probe family and frame regime that generated it.

## 6.3 Incomparable Thresholds

Distinct frame demands may generate incomparable thresholds.

**Proposition 6.11** (Incomparable Thresholds Exist)**.** *There exist a finite comparison artifact and frame regimes*

$$\varphi, \psi \in \mathrm{Eq}(\mathcal{E})$$

*such that*

$$\Theta_P(\varphi)$$

*and*

$$\Theta_P(\psi)$$

*are incomparable under refinement.*

*Proof.* Let

$$\mathcal{E} = \{F_0, F_1, F_2\}, \qquad \mathcal{A} = \{a, b, c\}, \qquad P = \{p\}.$$

Define the interpretation by

$$I(p, F_0) = a, \qquad I(p, F_1) = b, \qquad I(p, F_2) = c.$$

Let $\varphi$ induce the frame partition

$$\mathcal{E}/\varphi = \big\{\{F_0, F_1\}, \{F_2\}\big\},$$

and let $\psi$ induce

$$\mathcal{E}/\psi = \big\{\{F_0, F_2\}, \{F_1\}\big\}.$$

The threshold generated by $\varphi$ has occurrence partition

$$\mathcal{A}/\Theta_P(\varphi) = \big\{\{a, b\}, \{c\}\big\}.$$

The threshold generated by $\psi$ has occurrence partition

$$\mathcal{A}/\Theta_P(\psi) = \big\{\{a, c\}, \{b\}\big\}.$$

The first threshold relates $a$ and $b$ but not $a$ and $c$. The second relates $a$ and $c$ but

not $a$ and $b$.

Neither relation contains the other, so the thresholds are incomparable. □

The example exhibits two different occurrence demands. One requires the outputs $a$ and $b$ to be identified. The other requires $a$ and $c$ to be identified. Neither demand is uniformly stronger than the other.

When both frame demands must be supported, their thresholds combine by join.

**Theorem 6.12** (Threshold of Combined Frame Demands)**.** *Let*

$$\{\varphi_i\}_{i\in J}$$

*be a finite family of frame regimes. Then*

$$\Theta_P\left(\bigvee_{i\in J}\varphi_i\right)=\bigvee_{i\in J}\Theta_P(\varphi_i).$$

*Proof.* By Definition 6.4, the threshold map is $L_P$. The result is therefore Theorem 5.12. □

**Corollary 6.13** (Simultaneous Admissibility)**.** *An occurrence regime $\rho$ is admissible for every $\varphi_i$ if and only if*

$$\bigvee_{i\in J}\Theta_P(\varphi_i)\sqsubseteq\rho.$$

*Proof.* The regime $\rho$ is admissible for every $\varphi_i$ when

$$\Theta_P(\varphi_i)\sqsubseteq\rho$$

for every $i\in J$.

By the defining property of join, this holds when

$$\bigvee_{i\in J}\Theta_P(\varphi_i)\sqsubseteq\rho.$$

□

In the construction of Proposition 6.11, the join of the two thresholds is

$$\top_{\mathcal{A}},$$

because the required pairs connect $a$ to both $b$ and $c$. Supporting both frame demands therefore requires all three occurrence environments to be equivalent.

The individual demands are not linearly ordered, but their joint threshold remains well defined.

## 6.4 Extremal Admissible Regimes

The refinement order reverses a possible informal reading of the words *least* and *greatest*.

The least element of an admissible set is its finest regime. The greatest element is its coarsest regime.

**Theorem 6.14** (Extremal Admissible Regimes)**.** *For every frame regime $\varphi$, the admissible set*

$$\mathrm{Adm}_P(\varphi)$$

*is nonempty and has:*

*(a) the least element*

$$\Theta_P(\varphi);$$

*(b) the greatest element*

$$\top_{\mathcal{A}}.$$

*Proof.* Corollary 6.6 shows that

$$\Theta_P(\varphi)$$

is admissible.

Corollary 6.7 shows that it refines every admissible regime, so it is the least element.

By Corollary 4.17,

$$K_P(\top_{\mathcal{A}}) = \top_{\mathcal{E}}.$$

Every frame regime satisfies

$$\varphi \sqsubseteq \top_{\mathcal{E}},$$

so

$$\top_{\mathcal{A}} \in \mathrm{Adm}_P(\varphi).$$

Since

$$\top_{\mathcal{A}}$$

is the greatest occurrence regime, it is the greatest admissible regime. □

The greatest admissible regime is therefore always available, but carries little diagnostic information. It identifies every pair of occurrence environments and consequently supports every frame regime.

The mathematically informative boundary is the least admissible regime, because it is the finest occurrence comparison that still supports the declared frame demand.

**Proposition 6.15** (Meet Closure of Admissibility)**.** *Let*

$$\{\rho_i\}_{i\in J} \subseteq \mathrm{Adm}_P(\varphi)$$

*be a nonempty family of admissible occurrence regimes. Then*

$$\bigwedge_{i\in J} \rho_i \in \mathrm{Adm}_P(\varphi).$$

*Proof.* Since each $\rho_i$ is admissible, Theorem 6.5 gives

$$\Theta_P(\varphi) \sqsubseteq \rho_i$$

for every $i \in J$.

Thus $\Theta_P(\varphi)$ is a lower bound of the family. By the defining property of meet,

$$\Theta_P(\varphi) \sqsubseteq \bigwedge_{i\in J} \rho_i.$$

Applying the threshold characterization again proves that the meet is admissible. □

**Corollary 6.16** (Threshold as the Meet of All Admissible Regimes)**.** *For every frame regime $\varphi$,*

$$\Theta_P(\varphi) = \bigwedge \mathrm{Adm}_P(\varphi).$$

*Proof.* The threshold belongs to the admissible set and refines every member of it. It is therefore the greatest lower bound of the admissible regimes. □

*Remark* 6.17 (The Existence Question Is Resolved). Before the adjoint construction is available, one may ask whether the admissible occurrence regimes are closed under meet and therefore possess a finest member.

The adjunction gives the stronger result. The admissible set is not merely meet-closed. It is the principal upper set generated by

$$L_P(\varphi).$$

No additional closure hypothesis is required. The finest admissible occurrence regime exists for every frame regime in the finite comparison artifact and is

$$\Theta_P(\varphi) = L_P(\varphi).$$

Thresholds therefore summarize the exact occurrence-equivalence demand made by a frame regime under a fixed probe family.

Their values may be equal, strictly ordered, or incomparable. Multiple demands combine by join, and every admissible occurrence regime lies in the principal upper set above their combined threshold.

# 7 Representation

The threshold construction assigns an occurrence regime to an entire frame regime. The same construction can be applied to the least frame regime that identifies one selected pair of frame environments.

The resulting occurrence regime measures the exact output equivalences required to make that pair interpretively indistinguishable. These pairwise values determine the complete interpretive-kernel map.

## 7.1 Pairwise Interpretive Demand

**Definition 7.1** (Principal Pair Regime)**.** For frame environments $F, F' \in \mathcal{E}$, define

$$\gamma_{F,F'} = \mathrm{EqCl}_{\mathcal{E}}\big(\{(F, F')\}\big).$$

The relation $\gamma_{F,F'}$ is the least frame regime that identifies $F$ and $F'$.

When $F \neq F'$, the partition induced by $\gamma_{F,F'}$ has one block $\{F, F'\}$ and a singleton block for every other frame environment. When $F = F'$,

$$\gamma_{F,F} = \bot_{\mathcal{E}}.$$

**Definition 7.2** (Pairwise Interpretive Demand)**.** The *pairwise interpretive demand* induced by the probe family $P$ is the function

$$\delta_P : \mathcal{E} \times \mathcal{E} \longrightarrow \mathrm{Eq}(\mathcal{A})$$

defined by

$$\delta_P(F, F') = L_P\big(\gamma_{F,F'}\big).$$

The value $\delta_P(F, F')$ is the least occurrence regime under which the probes in $P$ make $F$ and $F'$ interpretively equivalent.

**Proposition 7.3** (Explicit Pairwise Construction)**.** *For every $F, F' \in \mathcal{E}$,*

$$\delta_P(F, F') = \mathrm{EqCl}_{\mathcal{A}}\Big(\big\{\big(I(p, F), I(p, F')\big) \;\big|\; p \in P\big\}\Big).$$

*Proof.* By Definition 7.2,

$$\delta_P(F, F') = L_P\big(\gamma_{F,F'}\big).$$

The principal pair regime is generated by the single pair $(F, F')$. Applying Theorem 5.6 to that generating relation gives

$$L_P\big(\gamma_{F,F'}\big) = \mathrm{EqCl}_{\mathcal{A}}\Big(\big\{\big(I(p, F), I(p, F')\big) \;\big|\; p \in P\big\}\Big).$$

□

Thus the pairwise demand may be computed directly. For each probe, the two interpreted outputs contribute one generating pair, and equivalence closure produces the least occurrence regime containing all of them.

**Theorem 7.4** (Pairwise Threshold Characterization)**.** *For every occurrence regime*

$$\rho \in \mathrm{Eq}(\mathcal{A})$$

*and frame environments* $F, F' \in \mathcal{E}$*,*

$$F \; K_P(\rho) \; F' \quad \Longleftrightarrow \quad \delta_P(F, F') \sqsubseteq \rho.$$

*Proof.* Because $K_P(\rho)$ is an equivalence relation,

$$F \; K_P(\rho) \; F'$$

holds when

$$\gamma_{F,F'} \sqsubseteq K_P(\rho).$$

By Theorem 5.8, this is equivalent to

$$L_P\big(\gamma_{F,F'}\big) \sqsubseteq \rho.$$

Definition 7.2 identifies the left-hand side as $\delta_P(F, F')$. □

The complete kernel can therefore be recovered from the pairwise demands:

$$K_P(\rho) = \big\{(F, F') \in \mathcal{E} \times \mathcal{E} \;\big|\; \delta_P(F, F') \sqsubseteq \rho\big\}.$$

## 7.2 Generalized Ultrametric Laws

The codomain of $\delta_P$ is the partition lattice

$$\mathrm{Eq}(\mathcal{A}).$$

The lattice join plays the role taken by maximum in an ordinary ultrametric.

**Definition 7.5** (Equivalence-Valued Generalized Ultrametric)**.** Let $X$ and $A$ be sets. An $\mathrm{Eq}(A)$*-valued generalized ultrametric* on $X$ is a function

$$\delta : X \times X \longrightarrow \mathrm{Eq}(A)$$

satisfying:

(U1) *reflexive zero:*

$$\delta(x, x) = \bot_A;$$

(U2) *symmetry:*

$$\delta(x, y) = \delta(y, x);$$

(U3) *strong triangle inequality:*

$$\delta(x, z) \sqsubseteq \delta(x, y) \vee \delta(y, z).$$

The term *zero* refers to the least equivalence relation, not to a numerical value.

The strong triangle inequality says that every occurrence equivalence required to identify $x$ with $z$ is contained in the regime generated jointly by the demands for $x$ with $y$ and $y$ with $z$.

**Theorem 7.6** (Interpretive Demand Is a Generalized Ultrametric)**.** *The pairwise interpretive demand*

$$\delta_P : \mathcal{E} \times \mathcal{E} \longrightarrow \mathrm{Eq}(\mathcal{A})$$

*is an* $\mathrm{Eq}(\mathcal{A})$*-valued generalized ultrametric.*

*Proof.* For reflexive zero,

$$\gamma_{F,F} = \bot_{\mathcal{E}}.$$

By Corollary 5.13,

$$\delta_P(F, F) = L_P\big(\bot_{\mathcal{E}}\big) = \bot_{\mathcal{A}}.$$

For symmetry,

$$\gamma_{F,F'} = \gamma_{F',F},$$

so

$$\delta_P(F, F') = \delta_P(F', F).$$

For the strong triangle inequality, let

$$\rho = \delta_P(F, F') \vee \delta_P(F', F'').$$

Since

$$\delta_P(F, F') \sqsubseteq \rho,$$

Theorem 7.4 gives

$$F \; K_P(\rho) \; F'.$$

Similarly,

$$F' \; K_P(\rho) \; F''.$$

The relation $K_P(\rho)$ is transitive by Proposition 4.3, so

$$F \; K_P(\rho) \; F''.$$

Applying Theorem 7.4 again gives

$$\delta_P(F, F'') \sqsubseteq \rho.$$

Substituting the definition of $\rho$ yields

$$\delta_P(F, F'') \sqsubseteq \delta_P(F, F') \vee \delta_P(F', F'').$$

□

*Remark* 7.7 (Distinct Frames May Have Zero Demand). The generalized ultrametric need not separate distinct frame environments.

For $F \neq F'$,

$$\delta_P(F, F') = \bot_{\mathcal{A}}$$

when

$$I(p, F) = I(p, F')$$

for every $p \in P$.

Distinct frame representations may therefore have zero pairwise demand when the

selected probes produce identical canonical occurrence environments.

The construction is consequently a generalized pseudoultrametric when the word *ultrametric* is reserved for separating distances. No separation hypothesis is needed for the representation result below.

## 7.3 Representation Theorem

The preceding construction is not specific to interpretation functions. It characterizes all complete meet-preserving maps between finite partition lattices.

**Theorem 7.8** (Representation of Complete Meet Homomorphisms)**.** *Let $A$ and $X$ be finite sets, and let*

$$K : \mathrm{Eq}(A) \longrightarrow \mathrm{Eq}(X)$$

*be a map.*

*The following are equivalent:*

*(a) $K$ preserves all meets, including the empty meet;*

*(b) there exists an $\mathrm{Eq}(A)$-valued generalized ultrametric*

$$\delta : X \times X \longrightarrow \mathrm{Eq}(A)$$

*such that for every $\rho \in \mathrm{Eq}(A)$ and $x, y \in X$,*

$$x \; K(\rho) \; y \quad \Longleftrightarrow \quad \delta(x, y) \sqsubseteq \rho.$$

*When it exists, the representing generalized ultrametric is unique and is given by*

$$\delta_K(x, y) = \bigwedge \{ \rho \in \mathrm{Eq}(A) \mid x \; K(\rho) \; y \}.$$

*Proof.* Suppose first that $K$ preserves all meets, including the empty meet.

Empty-meet preservation gives

$$K(\top_A) = \top_X.$$

Therefore, for every $x, y \in X$, the set

$$S_{x,y} = \{\rho \in \mathrm{Eq}(A) \mid x \; K(\rho) \; y\}$$

is nonempty.

Define

$$\delta_K(x,y) = \bigwedge S_{x,y}.$$

Since $K$ preserves meets,

$$K\big(\delta_K(x,y)\big) = \bigwedge_{\rho \in S_{x,y}} K(\rho).$$

The pair $(x,y)$ belongs to every relation in the meet, so

$$x \; K\left(\delta_K(x,y)\right) \; y.$$

If

$$\delta_K(x,y) \sqsubseteq \rho,$$

monotonicity of $K$ gives

$$K\big(\delta_K(x,y)\big) \sqsubseteq K(\rho).$$

Hence

$$x \; K(\rho) \; y.$$

Conversely, if

$$x \; K(\rho) \; y,$$

then

$$\rho \in S_{x,y},$$

and therefore

$$\delta_K(x,y) \sqsubseteq \rho.$$

It remains to verify the generalized ultrametric laws.

Reflexivity of every $K(\rho)$ gives

$$S_{x,x} = \mathrm{Eq}(A),$$

so

$$\delta_K(x, x) = \bot_A.$$

Symmetry of every $K(\rho)$ gives

$$S_{x,y} = S_{y,x},$$

and hence

$$\delta_K(x, y) = \delta_K(y, x).$$

Let

$$\rho = \delta_K(x, y) \vee \delta_K(y, z).$$

The threshold characterization already proved gives

$$x\ K(\rho)\ y \qquad \text{and} \qquad y\ K(\rho)\ z.$$

Transitivity of $K(\rho)$ gives

$$x\ K(\rho)\ z.$$

Therefore

$$\delta_K(x, z) \sqsubseteq \rho,$$

which is the strong triangle inequality.

Conversely, suppose an $\mathrm{Eq}(A)$-valued generalized ultrametric $\delta$ is given. Define

$$K_\delta(\rho) = \big\{(x, y) \in X \times X \;\big|\; \delta(x, y) \sqsubseteq \rho\big\}.$$

Reflexive zero gives reflexivity, and symmetry of $\delta$ gives symmetry of $K_\delta(\rho)$.

If

$$\delta(x, y) \sqsubseteq \rho \qquad \text{and} \qquad \delta(y, z) \sqsubseteq \rho,$$

then

$$\delta(x, y) \vee \delta(y, z) \sqsubseteq \rho.$$

The strong triangle inequality gives

$$\delta(x, z) \sqsubseteq \rho.$$

Thus $K_\delta(\rho)$ is transitive and therefore an equivalence relation.

Finally, for any family

$$\{\rho_i\}_{i\in J},$$

$$\delta(x,y) \sqsubseteq \bigwedge_{i\in J} \rho_i$$

holds when

$$\delta(x,y) \sqsubseteq \rho_i$$

for every $i \in J$. Hence

$$K_\delta\left(\bigwedge_{i\in J} \rho_i\right) = \bigwedge_{i\in J} K_\delta(\rho_i).$$

For the empty family, the right-hand condition is vacuous, so

$$K_\delta(\top_A) = \top_X.$$

Uniqueness follows because any representing generalized ultrametric must satisfy

$$\delta(x,y) = \bigwedge\{\rho \in \mathrm{Eq}(A) \mid x\ K(\rho)\ y\}.$$

□

**Corollary 7.9** (Representation of Interpretive Kernels)**.** *For every finite probe family $P$, the complete meet homomorphism*

$$K_P : \mathrm{Eq}(\mathcal{A}) \longrightarrow \mathrm{Eq}(\mathcal{E})$$

*is represented by the unique generalized ultrametric*

$$\delta_P(F,F') = L_P\big(\gamma_{F,F'}\big).$$

*In particular,*

$$K_P(\rho) = \big\{(F,F') \in \mathcal{E}\times\mathcal{E} \mid \delta_P(F,F') \sqsubseteq \rho\big\}.$$

The representation converts a lattice homomorphism into a pairwise threshold function. Conversely, the pairwise threshold function determines the entire family of kernel equivalence relations.

## 7.4 Exact Hypotheses and Limits

The representation theorem depends on complete meet preservation, not on monotonicity alone.

A monotone map

$$K : \mathrm{Eq}(A) \longrightarrow \mathrm{Eq}(X)$$

need not be recoverable from least pairwise thresholds. For a pairwise threshold to represent $K$, the pair must belong to $K$ at the meet of every regime under which it belongs. That is the closure property supplied by meet preservation.

*Remark* 7.10 (Empty-Meet Preservation). Preservation of the empty meet is the condition

$$K(\top_A) = \top_X.$$

It ensures that every pair $x, y \in X$ is equivalent under at least one occurrence regime, so the defining set for $\delta_K(x, y)$ is nonempty.

If only nonempty meets are assumed, top preservation must be imposed separately.

*Remark* 7.11 (No Join-Preservation Claim). The representation theorem does not imply

$$K(\rho \vee \sigma) = K(\rho) \vee K(\sigma).$$

A right adjoint preserves meets. It need not preserve joins.

The strong triangle inequality uses join inside the value lattice, but does not convert $K$ into a join homomorphism.

*Remark* 7.12 (Canonicalization Remains Prior). The representation theorem begins with a map whose domain is already $\mathrm{Eq}(\mathcal{A})$.

It does not choose a canonical occurrence representation, construct a comparison lifting, or justify quotienting over alternative matchings. Those choices determine the carrier and occurrence regimes to which the theorem applies.

A proposed comparison that fails meet preservation is not repaired by the representation theorem. It fails the theorem's hypothesis.

*Remark* 7.13 (No Scalar Ranking). The values of $\delta_P$ are equivalence relations and may be incomparable under refinement.

The construction therefore supplies no canonical scalar distance, total ranking, or single numerical severity measure. Any scalarization would add structure not contained in the partition lattice.

*Remark* 7.14 (Finite and Infinite Forms). The representation is not intrinsically finite. For any set $A$, the equivalence relations $\mathrm{Eq}(A)$ form a complete lattice, so the meets used in the proof of Theorem 7.8—the pairwise value $\bigwedge S_{x,y}$ and the arbitrary meets $\bigwedge_i \rho_i$—exist without any finiteness assumption. The representation and its uniqueness therefore hold for arbitrary $A$ and $X$.

Finiteness is required only by the computational claims of Section 10: interned occurrence identifiers, hashed signature grouping, generating-forest thresholds, finite witness extraction, and the executable schedule realization.

*Remark* 7.15 (Observational Rather Than Ontological Distance). The value

$$\delta_P(F, F')$$

does not measure how different the frame environments are in themselves.

It records the least occurrence regime that makes them indistinguishable under the selected probes and interpretation function.

Changing $P$, changing $I$, changing the canonical occurrence carrier, or changing the comparison lifting may change the value without changing either frame representation.

The representation result therefore characterizes the algebra carried by interpretive kernels without claiming that every proposed output comparison produces such a kernel.

The next section supplies finite counterexamples that separate the valid construction from comparison rules that lose canonical structure or fail the required preservation laws.

# 8 Finite Counterexamples

The preceding results depend on three structural choices.

First, occurrence environments are placed in a declared canonical representation

before they are compared. Second, occurrence and frame regimes are required to be equivalence relations. Third, the interpretive-kernel map preserves meets because comparison is performed pointwise over a fixed probe family.

These requirements are not presentational conveniences. This section gives finite counterexamples to several natural alternatives.

The first counterexample shows that index-wise comparison is not well defined on unordered output families without canonicalization.

The second rejects a proposed repair that minimizes over all bijections. That repair is permutation-invariant, but it does not preserve meets and its associated lattice-valued comparison fails the strong triangle inequality.

The third separates resolution policies from frame identity regimes. A deterministic fallback procedure need not induce an equivalence relation at all.

## 8.1 Presentation Dependence Without Canonicalization

Suppose raw interpretation produces an unordered family of two occurrence values drawn from a finite carrier

$$O = \{a, b\}.$$

The unordered family

$$X = \{a, b\}$$

has both ordered presentations

$$(a, b) \qquad \text{and} \qquad (b, a).$$

Let equality be the component comparison. Consider the attempted index-wise rule

$$(x_1, x_2) \mathrel{\widehat{=}} (y_1, y_2)$$

when

$$x_1 = y_1 \qquad \text{and} \qquad x_2 = y_2.$$

Using the same presentation on both sides gives

$$(a, b) \mathrel{\widehat{=}} (a, b).$$

Using two different presentations of the same unordered family gives

$$(a, b) \mathrel{\widehat{\neq}} (b, a).$$

The attempted relation therefore assigns different comparison results to different representatives of the same unordered object.

**Proposition 8.1** (Index-Wise Comparison Does Not Descend)**.** *Index-wise component comparison does not define a relation on unordered finite families unless a representative is selected by a declared canonicalization or the comparison is otherwise proved independent of representative choice.*

*Proof.* The family

$$X = \{a, b\}$$

is equal to itself as an unordered family.

The ordered representatives

$$(a, b) \qquad \text{and} \qquad (a, b)$$

satisfy index-wise equality, while the ordered representatives

$$(a, b) \qquad \text{and} \qquad (b, a)$$

do not.

The comparison therefore depends on the selected representatives and does not descend to the quotient by permutation. □

A declared canonicalization removes the ambiguity. If a declared order places $a$ before $b$, then both presentations are mapped to

$$(a, b),$$

and comparison is performed on that canonical indexed form.

The construction does not require that every domain use ordering. It requires that the domain declare how an occurrence environment is represented before index-wise comparison is applied.

## 8.2 Failure of the Matching Comparison

A natural response to Proposition 8.1 is to avoid selecting a canonical order and instead compare unordered families through possible bijections.

Let $O$ be a finite component carrier, and let

$$\rho \in \mathrm{Eq}(O).$$

For equal-cardinality finite families $X$ and $Y$, define the proposed matching lift

$$X \mathrel{\widehat{\rho}_{\mathrm{m}}} Y$$

when there exists a bijection

$$\sigma : X \longrightarrow Y$$

such that

$$x \mathrel{\rho} \sigma(x) \qquad \text{for every } x \in X.$$

The definition is invariant under permutation. It nevertheless fails to preserve meets.

For compactness, write a partition by listing its blocks. Thus

$$ac|bd$$

denotes

$$\big\{\{a, c\}, \{b, d\}\big\}.$$

**Proposition 8.2** (Matching Lift Does Not Preserve Meets)**.** *There exist component equivalences*

$$\rho, \sigma \in \mathrm{Eq}(O)$$

*and unordered finite families $X, Y$ such that*

$$X \mathrel{\widehat{\rho}_{\mathrm{m}}} Y \qquad \textit{and} \qquad X \mathrel{\widehat{\sigma}_{\mathrm{m}}} Y,$$

*but*

$$X \not\mathrel{\widehat{\rho \wedge \sigma}_{\mathrm{m}}} Y.$$

*Consequently,*

$$\widehat{\rho \wedge \sigma}_{\mathrm{m}} \neq \widehat{\rho}_{\mathrm{m}} \wedge \widehat{\sigma}_{\mathrm{m}}.$$

*Proof.* Let

$$O = \{a, b, c, d\},$$

and define

$$\rho = ac|bd, \qquad \sigma = ad|bc.$$

Let

$$X = \{a, b\}, \qquad Y = \{c, d\}.$$

Under $\rho$, the bijection

$$a \longmapsto c, \qquad b \longmapsto d$$

witnesses

$$X \mathrel{\widehat{\rho}_{\mathrm{m}}} Y.$$

Under $\sigma$, the different bijection

$$a \longmapsto d, \qquad b \longmapsto c$$

witnesses

$$X \mathrel{\widehat{\sigma}_{\mathrm{m}}} Y.$$

The meet of the component equivalences is

$$\rho \wedge \sigma = a|b|c|d,$$

the equality relation on $O$.

The families $X$ and $Y$ are disjoint, so no bijection between them pairs equal

component values. Therefore

$$X \not\sim_{\widehat{\rho \wedge \sigma}_{\mathrm{m}}} Y.$$

□

The failure occurs because the bijection witnessing comparison under $\rho$ need not be the bijection witnessing comparison under $\sigma$. Membership in the meet of the lifted relations permits separate witnesses, while lifting the meet requires one witness that satisfies both component regimes simultaneously.

Permutation invariance therefore does not imply meet preservation.

The same obstruction appears in the lattice-valued version of the proposal.

Let

$$d : O \times O \longrightarrow \mathrm{Eq}(O)$$

be defined by

$$d(u, v) = \mathrm{EqCl}_O\big(\{(u, v)\}\big).$$

Thus $d(u, v)$ is the least component equivalence that identifies $u$ and $v$.

**Lemma 8.3** (The Component Comparison Is a Generalized Ultrametric)**.** *The function $d$ is an $\mathrm{Eq}(O)$-valued generalized ultrametric.*

*Proof.* For every $u \in O$,

$$d(u, u) = \bot_O.$$

Symmetry follows because the least equivalence relation containing $(u, v)$ is also the least equivalence relation containing $(v, u)$.

For $u, v, w \in O$, the join

$$d(u, v) \vee d(v, w)$$

contains both pairs $(u, v)$ and $(v, w)$. By transitivity, it contains $(u, w)$. Therefore

$$d(u, w) \sqsubseteq d(u, v) \vee d(v, w).$$

□

For equal-cardinality unordered families, define the proposed best-matching com-

parison

$$d_{\mathrm{m}}(X,Y) = \bigwedge_{\sigma : X \cong Y} \bigvee_{x \in X} d\big(x, \sigma(x)\big).$$

The inner join is the occurrence regime required by one selected matching. The outer meet attempts to retain only the equivalences common to all possible matching costs.

**Proposition 8.4** (Best-Matching Comparison Fails the Strong Triangle Law)**.** *The comparison $d_{\mathrm{m}}$ need not satisfy*

$$d_{\mathrm{m}}(X,Z) \sqsubseteq d_{\mathrm{m}}(X,Y) \vee d_{\mathrm{m}}(Y,Z).$$

*Proof.* Let

$$O = \{a, b, c, d\},$$

and take

$$X = \{a, b\}, \qquad Y = \{a, c\}, \qquad Z = \{b, d\}.$$

There are two bijections between each pair of families.

For $X$ and $Y$, the two matching costs are

$$a|bc|d \qquad \text{and} \qquad abc|d.$$

Their meet is

$$d_{\mathrm{m}}(X,Y) = a|bc|d.$$

For $Y$ and $Z$, the two matching costs are

$$ab|cd \qquad \text{and} \qquad ad|bc.$$

Their meet is

$$d_{\mathrm{m}}(Y,Z) = a|b|c|d = \bot_O.$$

For $X$ and $Z$, the two matching costs are

$$abd|c \qquad \text{and} \qquad ad|b|c.$$

Their meet is

$$d_{\mathrm{m}}(X, Z) = ad|b|c.$$

Therefore

$$d_{\mathrm{m}}(X, Y) \vee d_{\mathrm{m}}(Y, Z) = a|bc|d.$$

But

$$ad|b|c \not\sqsubseteq a|bc|d,$$

because the left-hand relation identifies $a$ with $d$, while the right-hand relation does not.

Hence

$$d_{\mathrm{m}}(X, Z) \not\sqsubseteq d_{\mathrm{m}}(X, Y) \vee d_{\mathrm{m}}(Y, Z).$$

□

The proposition gives an immediate failure at the level of threshold relations.

Let

$$\rho = a|bc|d,$$

and define

$$X \; R_\rho \; Y \quad \Longleftrightarrow \quad d_{\mathrm{m}}(X, Y) \sqsubseteq \rho.$$

Then

$$X \; R_\rho \; Y$$

and

$$Y \; R_\rho \; Z,$$

but

$$X \; \not R_\rho \; Z.$$

Thus $R_\rho$ is not transitive. The best-matching comparison does not merely fail to produce the desired adjunction. Its threshold relations need not be frame regimes at all.

*Remark* 8.5 (Source of the Failure). For any fixed pair of bijections, composition and the component strong triangle law provide the expected bound.

The invalid step is moving the meets over alternative bijections through the lattice join.

That step would require a distributive law not available in $\mathrm{Eq}(O)$, whose partition lattice is generally nondistributive.

The canonicalized construction avoids this step. After canonicalization, one indexed comparison is made. There is no meet over alternative matchings, and arbitrary meets of component regimes commute with the universal index-wise condition.

## 8.3 Resolution Policies Are Not Frame Regimes

A frame regime is an equivalence relation on the frame carrier. A resolution policy instead selects which frame will be used when a declaration is evaluated.

The two objects may interact, but are not interchangeable.

Consider three frame releases

$$r_0, r_1, r_2,$$

ordered from oldest to newest.

Suppose a pin-with-fallback policy behaves as follows:

(a) a declaration pinned to $r_i$ resolves to $r_i$ when that release is available;
(b) when $r_2$ is unavailable, the declaration may resolve to $r_1$;
(c) when $r_1$ is unavailable, the declaration may resolve to $r_0$.

If this policy is misread as a frame-sameness relation, the resulting directed relation is

$$R_{\mathrm{pf}} = \bot_{\mathcal{E}} \cup \{(r_2, r_1), (r_1, r_0)\}.$$

**Proposition 8.6** (Pin-With-Fallback Need Not Be a Frame Regime)**.** *The relation $R_{\mathrm{pf}}$ is not an equivalence relation.*

*Proof.* The relation contains

$$(r_2, r_1),$$

but does not contain

$$(r_1, r_2).$$

It is therefore not symmetric.

In the displayed finite policy, it also contains

$$(r_2, r_1) \qquad \text{and} \qquad (r_1, r_0),$$

but not

$$(r_2, r_0),$$

so it is not transitive either. □

The failure is not repaired by requiring the policy to be deterministic. A total deterministic function

$$q : D \times C \longrightarrow \mathcal{E}$$

from declarations and availability contexts to selected frames is still a selection function, not an equivalence relation on $\mathcal{E}$.

The same distinction applies to always-latest, latest-compatible, and similar deployment practices. They answer *which frame should be selected?* rather than *which frames are the same?*

**Corollary 8.7** (Partition Validation Is Required)**.** *A declared frame regime must be validated as an equivalence relation before it is supplied to $L_P$, $K_P$, or the threshold construction.*

*A resolution policy that fails this validation may be represented as part of the frame environment or interpretation procedure, but not as a frame regime.*

## 8.4 Further Non-Implications

The preceding examples delimit several additional claims that do not follow from the valid construction.

**Proposition 8.8** (Monotonicity Does Not Imply Meet Preservation)**.** *A monotone map between finite partition lattices need not preserve meets, even when it preserves the greatest element.*

*Proof.* Let

$$A = \{a, b, c\}, \qquad X = \{x, y\}.$$

Define

$$\alpha = ab|c, \qquad \beta = ac|b.$$

Define

$$M : \mathrm{Eq}(A) \longrightarrow \mathrm{Eq}(X)$$

by

$$M(\rho) = \begin{cases} \top_X, & \text{when } \alpha \sqsubseteq \rho \text{ or } \beta \sqsubseteq \rho; \\ \bot_X, & \text{otherwise.} \end{cases}$$

The map is monotone, and

$$M(\top_A) = \top_X.$$

However,

$$\alpha \wedge \beta = \bot_A,$$

so

$$M\big(\alpha \wedge \beta\big) = \bot_X.$$

On the other hand,

$$M(\alpha) = M(\beta) = \top_X,$$

and therefore

$$M(\alpha) \wedge M(\beta) = \top_X.$$

Hence

$$M\big(\alpha \wedge \beta\big) \neq M(\alpha) \wedge M(\beta).$$

□

Monotonicity alone therefore does not supply the adjunction, least pairwise thresholds, or generalized-ultrametric representation. The stronger meet-preservation law is essential.

**Proposition 8.9** (Meet Preservation Does Not Imply Join Preservation)**.** *An interpretive-kernel map may preserve every meet while failing to preserve joins.*

*Proof.* Let

$$\mathcal{A} = \{a, b, c\}, \qquad \mathcal{E} = \{F_0, F_1\}, \qquad P = \{p\}.$$

Define

$$I(p, F_0) = a, \qquad I(p, F_1) = c.$$

Let

$$\rho = ab|c, \qquad \sigma = a|bc.$$

Neither occurrence regime identifies $a$ with $c$. Therefore

$$K_P(\rho) = K_P(\sigma) = \bot_{\mathcal{E}}.$$

Their join is

$$\rho \vee \sigma = \top_{\mathcal{A}},$$

because the pairs $a \sim b$ and $b \sim c$ generate $a \sim c$.

Consequently,

$$K_P\big(\rho \vee \sigma\big) = \top_{\mathcal{E}},$$

while

$$K_P(\rho) \vee K_P(\sigma) = \bot_{\mathcal{E}}.$$

Thus

$$K_P\big(\rho \vee \sigma\big) \neq K_P(\rho) \vee K_P(\sigma).$$

□

*Remark* 8.10 (Zero Demand Does Not Imply Frame Equality). Distinct frame environments may satisfy

$$\delta_P(F, F') = \bot_{\mathcal{A}}$$

when every probe produces the same canonical occurrence environment under both.

Zero pairwise demand establishes observational equivalence under $P$, not equality of frame representation.

*Remark* 8.11 (Admissibility Does Not Imply Minimality). An admissible occurrence regime need not equal the threshold.

Admissibility gives

$$\Theta_P(\varphi) \sqsubseteq \rho.$$

Equality holds only when $\rho$ introduces no occurrence equivalence beyond those required by the frame demand and probe family.

*Remark* 8.12 (Permutation Invariance Is Insufficient). The matching comparison is permutation-invariant, but fails both meet preservation and the generalized ultrametric law.

Permutation invariance therefore does not by itself make an unordered comparison suitable for interpretive-kernel construction.

The counterexamples identify the boundary of the valid construction.

A domain may use unordered raw output, resolution policies, or comparison procedures other than index-wise equality. It must nevertheless supply:

(a) a declared canonical occurrence carrier;
(b) occurrence and frame comparisons that are genuine equivalence relations;
(c) a comparison lifting satisfying the required preservation laws; and
(d) both the probe family and occurrence regime indexing every reported kernel.

The schedule realization of the next section meets these requirements through chronological canonicalization, finite probe horizons, declared frame-environment tuples, and index-wise occurrence comparison.

# 9 Schedule Interpretation

This section instantiates the preceding construction for finite schedule interpretation.

The schedule realization preserves the distinction between semantic policy and external interpretive facts. Denotation-affecting policy belongs to the schedule declaration and has no implicit default. The frame environment supplies the cited external artifacts under which that declaration is interpreted.

A probe combines a schedule declaration with a finite observation horizon and a requested output channel. Interpretation produces a finite occurrence environment, which is placed in chronological canonical form before an occurrence regime is applied.

## 9.1 Declared Schedule Signature

Let

$$\mathcal{S}$$

be a finite carrier of well-formed schedule declarations.

A declaration contains the semantic choices that determine its meaning, including any applicable recurrence, exception, resolution, ambiguity, month-length, observance, or combination policy. A denotation-affecting policy is represented explicitly. It is not supplied by an undeclared implementation default.

Let

$$\mathcal{C}$$

be a finite carrier of civil occurrence values, and let

$$\mathcal{T}$$

be a finite carrier of resolved instants.

The two carriers represent distinct output dimensions.

A civil value records a position in a declared civil-time domain, including its date, local time, and zone-bearing type where applicable.

A resolved value records an instant on the resolved time line.

The tagged schedule-output carrier is

$$\mathcal{V} = \big(\{\mathsf{civil}\} \times \mathcal{C}\big) \ \cup \ \big(\{\mathsf{resolved}\} \times \mathcal{T}\big).$$

The tags prevent a civil value and a resolved instant from being treated as values of the same output dimension.

**Definition 9.1** (Schedule Probe)**.** A *schedule probe* is a tuple

$$p = \langle s, W, \chi \rangle,$$

where:

(a) $s \in \mathcal{S}$ is a well-formed schedule declaration;

(b) $W$ is a finite observation horizon; and

(c)
$$\chi \in \{\mathsf{civil}, \mathsf{resolved}\}$$
is the requested output channel.

A probe requests one typed view of the schedule's occurrences. Two probes may use the same declaration and horizon while requesting different output channels.

Let
$$\mathsf{End} = \{\mathsf{exhausted}, \mathsf{bounded}\}.$$

The value exhausted records that the interpreted declaration produced no further occurrence because its denotation ended within the horizon.

The value bounded records that interpretation stopped at the declared probe boundary.

**Definition 9.2** (Schedule Occurrence Environment)**.** A *schedule occurrence environment* is a pair
$$A = \langle \mathbf{v}, e \rangle,$$
where
$$\mathbf{v} = \langle v_1, \ldots, v_n \rangle$$
is a finite canonical sequence of values from $\mathcal{V}$, and
$$e \in \mathsf{End}$$
is its terminal tag.

Let
$$\mathcal{A}_{\mathrm{sch}}$$
be the finite carrier of schedule occurrence environments admitted to the comparison artifact.

The schedule interpretation function has signature
$$I_{\mathrm{sch}} : P \times \mathcal{E}_{\mathrm{sch}} \longrightarrow \mathcal{A}_{\mathrm{sch}}.$$

For

$$p = \langle s, W, \chi \rangle$$

and frame environment $F$, the value

$$I_{\mathrm{sch}}(p, F)$$

is the canonical finite sequence obtained by interpreting $s$ under $F$ over $W$, projecting the result to channel $\chi$, and attaching the appropriate terminal tag.

## 9.2 Chronological Canonicalization

A schedule may generate occurrences through unions, intersections, exceptions, observance rules, recurrence expansion, or other compositional operators.

The order in which an implementation discovers those occurrences is not part of their meaning unless the schedule declaration explicitly makes it so. Comparison therefore does not use generation order.

**Definition 9.3** (Schedule Canonicalization)**.** Schedule canonicalization maps a finite raw family of typed occurrence values to a finite sequence ordered:

(a) primarily by chronological position in the requested output channel; and
(b) secondarily by a declared canonical ordering of the complete typed occurrence representation.

Multiplicity is retained. The terminal tag is appended after ordering.

For the resolved channel, chronological order is order by instant.

For the civil channel, chronological order is the declared order of the civil-time type, including its date, local time, and zone-bearing information.

The secondary order makes canonicalization deterministic when distinct occurrence representations share the same primary chronological position.

*Remark* 9.4 (Canonicalization and Normalization). Canonicalization orders a schedule's represented occurrences. It does not silently change their denotation.

A domain normalization that alters, removes, combines, shifts, or resolves occurrences is a semantic operation and must be declared in the schedule expression or its explicit

policy.

Such an operation occurs before canonicalization.

*Remark* 9.5 (No Occurrence Matching). Two schedule occurrence environments are not compared by searching for a favorable bijection between their generated values.

Each is first placed in chronological canonical form. The declared occurrence regime is then applied to the resulting indexed sequences, including their lengths, multiplicities, and terminal tags.

## 9.3 Finite Recurrence Horizons

A recurring schedule may describe an indefinitely extensible denotation. That does not make a finite probe or its occurrence environment infinite.

**Definition 9.6** (Observation Horizon)**.** An *observation horizon* is a finite declared interval

$$W = [t_0, t_1)$$

in the input domain appropriate to the schedule declaration.

Interpretation includes the occurrences selected by the declaration whose relevant positions lie within $W$, subject to its explicit boundary policy.

The horizon belongs to the probe, not to the schedule declaration's identity. Changing the horizon changes the observation and therefore produces a different probe.

**Proposition 9.7** (Finite Schedule Output)**.** *Suppose:*

*(a) the schedule declaration is well formed;*
*(b) its operators are finite over every finite horizon;*
*(c) the frame environment resolves every cited external artifact; and*
*(d) the observation horizon is finite.*

*Then*

$$I_{\mathrm{sch}}(p, F)$$

*is a finite schedule occurrence environment.*

The proposition does not require the schedule declaration to denote a globally finite recurrence.

The sequence length and terminal tag remain part of the comparison. Two frames may produce the same canonical prefix while differing in whether the schedule exhausted or was merely cut off at the horizon.

*Remark* 9.8 (Horizon Relativity). A kernel computed under one horizon is not a claim about all future occurrences.

Extending the horizon may expose a frame distinction that was absent from the shorter probe family. In the abstract construction, this is ordinary probe-family extension.

## 9.4 Frame Environments

A schedule frame environment is a tuple of cited external artifacts used during interpretation.

**Definition 9.9** (Schedule Frame Environment)**.** A *schedule frame environment* is a tuple

$$F = \langle z, h, b, v \rangle,$$

where:

(a) $z$ identifies a time-zone rule artifact and its release;
(b) $h$ identifies a holiday-calendar artifact and its release;
(c) $b$ identifies a business-calendar artifact and its release; and
(d) $v$ identifies any provider or authority artifact whose cited facts participate in interpretation.

A concrete comparison artifact may fix some components across every frame and vary only the components under examination.

The frame tuple contains external facts. It does not absorb semantic policy that belongs in the schedule declaration.

For example, a holiday artifact may state which dates are holidays. Whether a schedule skips, shifts, retains, or otherwise treats those dates is schedule policy.

Likewise, a time-zone artifact supplies the applicable offset rules. A policy for resolving an ambiguous or nonexistent civil time belongs to the schedule declaration.

*Remark* 9.10 (Frame Regimes and Resolution Policies). A frame regime is an equivalence relation on schedule frame environments.

Exact pinning, pin-with-fallback, latest-compatible, and always-latest are resolution policies for selecting a frame environment. They are not frame regimes unless a separate declared equivalence relation is supplied and validated.

Frame regimes may compare complete tuples or may be induced by declared projections.

For example, the projection

$$q_z(F) = z$$

induces the frame regime

$$F \varphi_z F' \quad \Longleftrightarrow \quad q_z(F) = q_z(F').$$

Similarly,

$$q_h(F) = h$$

induces

$$F \varphi_h F' \quad \Longleftrightarrow \quad q_h(F) = q_h(F').$$

These regimes answer different questions. Two frame environments may share the same time-zone artifact while differing in holiday data, or share the same holiday artifact while differing in time-zone rules.

## 9.5 A Nondegenerate Finite Example

The following finite construction exhibits two independently varying frame components and two typed occurrence dimensions.

It is an abstract schedule example. A later worked application may replace the symbolic releases and outputs with specific cited artifacts, dates, and generated occurrences without changing the construction.

Let the frame carrier be

$$\mathcal{E}_{\mathrm{sch}} = \{F_0, F_1, F_2\},$$

where

$$F_0 = \langle z_0, h_0, b_0, v_0 \rangle,$$

$$F_1 = \langle z_1, h_0, b_0, v_0 \rangle,$$

and

$$F_2 = \langle z_0, h_1, b_0, v_0 \rangle.$$

Thus $F_1$ differs from $F_0$ only in its time-zone artifact, while $F_2$ differs from $F_0$ only in its holiday artifact.

Define the time-zone frame regime by equality of the time-zone projection:

$$\mathcal{E}_{\mathrm{sch}}/\varphi_z = \big\{\{F_0, F_2\}, \{F_1\}\big\}.$$

Define the holiday frame regime by equality of the holiday projection:

$$\mathcal{E}_{\mathrm{sch}}/\varphi_h = \big\{\{F_0, F_1\}, \{F_2\}\big\}.$$

The two frame regimes are non-discrete and incomparable.

Let the probe family be

$$P = \{p_z, p_h\}.$$

The probe $p_z$ requests resolved output for a local-time schedule whose instant differs between the time-zone artifacts $z_0$ and $z_1$ within the selected horizon.

The holiday artifacts do not affect this probe.

The probe $p_h$ requests civil output for a holiday-relative schedule whose civil occurrence differs between the holiday artifacts $h_0$ and $h_1$ within the selected horizon.

The time-zone artifacts do not affect this probe.

Let

$$u_0, u_1 \in \mathcal{T}, \qquad u_0 \neq u_1,$$

be the two resolved outputs, and let

$$c_0, c_1 \in \mathcal{C}, \qquad c_0 \neq c_1,$$

be the two civil outputs.

For compactness, write

$$A_{u_i} = \langle \langle (\mathsf{resolved}, u_i) \rangle , \mathsf{bounded} \rangle$$

and

$$A_{c_i} = \langle \langle (\mathsf{civil}, c_i) \rangle , \mathsf{bounded} \rangle .$$

The interpretation table is:

| | $p_z$ | $p_h$ |
|---|---|---|
| $F_0$ | $A_{u_0}$ | $A_{c_0}$ |
| $F_1$ | $A_{u_1}$ | $A_{c_0}$ |
| $F_2$ | $A_{u_0}$ | $A_{c_1}$ |

Let the examined occurrence carrier be

$$\mathcal{A}_0 = \{A_{u_0}, A_{u_1}, A_{c_0}, A_{c_1}\}.$$

Under strict equality on $\mathcal{A}_0$, the three frame environments have distinct probe signatures:

$$\Sigma_{P,\perp}(F_0) = \langle A_{u_0}, A_{c_0} \rangle,$$

$$\Sigma_{P,\perp}(F_1) = \langle A_{u_1}, A_{c_0} \rangle,$$

and

$$\Sigma_{P,\perp}(F_2) = \langle A_{u_0}, A_{c_1} \rangle.$$

Therefore

$$K_P\big(\perp_{\mathcal{A}_0}\big) = \perp_{\mathcal{E}_{\mathrm{sch}}}.$$

The threshold of the time-zone frame regime is

$$\Theta_P(\varphi_z) = A_{c_0} A_{c_1} \mid A_{u_0} \mid A_{u_1}.$$

Indeed, $\varphi_z$ requires $F_0$ and $F_2$ to remain equivalent. Their resolved outputs already agree, while their civil outputs differ. The least supporting occurrence regime therefore identifies $A_{c_0}$ with $A_{c_1}$ and introduces no other nontrivial equivalence.

Similarly, the threshold of the holiday frame regime is

$$\Theta_P(\varphi_h) = A_{u_0}A_{u_1} \mid A_{c_0} \mid A_{c_1}.$$

The regime $\varphi_h$ requires $F_0$ and $F_1$ to remain equivalent. Their civil outputs already agree, while their resolved outputs differ. The least supporting occurrence regime therefore identifies $A_{u_0}$ with $A_{u_1}$.

**Proposition 9.11** (Incomparable Schedule Thresholds)**.** *In the finite schedule construction above,*

$$\Theta_P(\varphi_z)$$

*and*

$$\Theta_P(\varphi_h)$$

*are non-discrete and incomparable.*

*Moreover,*

$$K_P\big(\Theta_P(\varphi_z)\big) = \varphi_z,$$

*and*

$$K_P\big(\Theta_P(\varphi_h)\big) = \varphi_h.$$

*Proof.* The threshold $\Theta_P(\varphi_z)$ identifies $A_{c_0}$ with $A_{c_1}$ but does not identify $A_{u_0}$ with $A_{u_1}$.

The threshold $\Theta_P(\varphi_h)$ identifies $A_{u_0}$ with $A_{u_1}$ but does not identify $A_{c_0}$ with $A_{c_1}$.

Neither relation contains the other, so they are incomparable.

Under $\Theta_P(\varphi_z)$, the signatures of $F_0$ and $F_2$ agree in both probe positions, while the signature of $F_1$ remains distinct through $p_z$. Hence

$$K_P\big(\Theta_P(\varphi_z)\big) = \varphi_z.$$

The argument for $\varphi_h$ is symmetric. □

The example is nondegenerate in four respects.

First, neither frame regime is equality or the indiscrete relation.

Second, neither threshold is equality or the indiscrete occurrence regime.

Third, the frame regimes are incomparable.

Fourth, their thresholds are incomparable because they require equivalence in different typed output dimensions.

Their combined demand satisfies

$$\Theta_P\big(\varphi_z \vee \varphi_h\big) = \Theta_P(\varphi_z) \vee \Theta_P(\varphi_h),$$

which has partition

$$A_{c_0} A_{c_1} \mid A_{u_0} A_{u_1}.$$

The civil and resolved channels remain distinct because the output carrier is typed. Combining the two frame demands does not force a civil occurrence value to be identified with a resolved instant.

This finite construction supplies the schedule-level shape required by the theory. The empirical application must additionally identify:

(a) the actual schedule declarations;
(b) the cited frame artifacts and releases;
(c) the finite horizons;
(d) the generated civil and resolved outputs;
(e) the declared frame regimes;
(f) the occurrence regimes admitted by the vocabulary; and
(g) the finite witnesses for every failed admissibility claim.

# 10 Computational Procedure

The constructions of the preceding sections are finite and executable.

A single table of canonical interpretation results supports computation of interpretive

kernels, generated occurrence regimes, thresholds, pairwise interpretive demands, admissibility verdicts, and finite witnesses.

The algebraic operations are inexpensive relative to schedule interpretation itself. The potentially costly step is generating and canonicalizing the occurrence environments. The procedure therefore separates interpretation cost from partition-processing cost and reuses each computed interpretation result.

## 10.1 Partition Construction

Let $X$ be a finite carrier.

A partition of $X$ is represented computationally by assigning a block label to every element. Write

$$\lambda_\alpha : X \longrightarrow B_\alpha$$

for a block-label function representing an equivalence relation

$$\alpha \in \mathrm{Eq}(X),$$

so that

$$x \; \alpha \; y \quad \Longleftrightarrow \quad \lambda_\alpha(x) = \lambda_\alpha(y).$$

The particular block identifiers are not significant. Only equality of labels is used.

**Definition 10.1** (Normalized Partition Representation)**.** A *normalized partition representation* of an equivalence relation $\alpha$ over a finite carrier $X$ consists of:

(a) a stable identifier for every element of $X$;
(b) one block identifier assigned to every element; and
(c) a block inventory containing the nonempty fibers of the block-label function.

Supplying disjoint nonempty blocks whose union is $X$ establishes partition-hood directly.

When a proposed regime is instead supplied as an arbitrary binary relation, the implementation must validate reflexivity, symmetry, and transitivity before admitting it as an occurrence or frame regime.

**Definition 10.2** (Interpretation Table)**.** For a finite comparison artifact, the *interpretation table* is

$$T_I : P \times \mathcal{E} \longrightarrow \mathcal{A}$$

with

$$T_I(p, F) = I(p, F).$$

Each table value is computed once, placed in canonical form, interned in the occurrence carrier, and thereafter referenced by its stable identifier.

The table contains

$$|P|\,|\mathcal{E}|$$

entries.

For schedule interpretation, constructing one entry includes recurrence generation, application of declared schedule policy, projection to the requested output channel, chronological canonicalization, and attachment of the terminal tag.

**Proposition 10.3** (Kernel Construction by Signatures)**.** *Let*

$$\rho \in \mathrm{Eq}(\mathcal{A})$$

*be represented by the block-label function* $\lambda_\rho$*.*

*For every frame environment* $F$*, construct the finite signature*

$$S_\rho(F) = \big(\lambda_\rho\big(T_I(p, F)\big)\big)_{p \in P}.$$

*Grouping frame environments by equality of these signatures produces the partition*

$$\mathcal{E}/K_P(\rho).$$

*Proof.* Two frame environments $F$ and $F'$ receive equal signatures when

$$\lambda_\rho\big(T_I(p, F)\big) = \lambda_\rho\big(T_I(p, F')\big)$$

for every $p \in P$.

Equality of the block labels holds when

$$I(p, F) \; \rho \; I(p, F')$$

for every $p \in P$.

By Definition 4.2, this condition is exactly

$$F \; K_P(\rho) \; F'.$$

□

A stable ordering of the probe identifiers may be used to serialize the signatures. The mathematical definition remains indexed by $P$ and does not depend on that implementation order.

Standard partition operations are also computable from block labels.

**Proposition 10.4** (Finite Partition Operations)**.** *Let*

$$\alpha, \beta \in \mathrm{Eq}(X)$$

*be represented by block labels.*

*Then:*

*(a) refinement*

$$\alpha \sqsubseteq \beta$$

*is decidable by checking that every $\alpha$-block lies within one $\beta$-block;*

*(b) the meet*

$$\alpha \wedge \beta$$

*is obtained by grouping each $x \in X$ under the paired label*

$$\big(\lambda_\alpha(x), \lambda_\beta(x)\big);$$

*(c) the join*

$$\alpha \vee \beta$$

*is obtained by initializing a union-find structure on $X$ and joining a spanning set of members within every block of $\alpha$ and every block of $\beta$.*

The join procedure does not enumerate all pairs within each block. A block with $k$ members requires only $k - 1$ union operations.

## 10.2 Generating Forests

Let

$$\varphi \in \mathrm{Eq}(\mathcal{E})$$

be a frame regime.

For each block

$$C \in \mathcal{E}/\varphi,$$

choose one member as its root and connect that root to every other member of $C$.

The union of those blockwise stars is a spanning forest

$$G_\varphi \subseteq \mathcal{E} \times \mathcal{E}$$

that generates $\varphi$.

Any other spanning tree within each block produces the same equivalence closure.

**Definition 10.5** (Forest-Based Generated-Regime Procedure)**.** To compute

$$L_P(\varphi),$$

perform the following steps:

(1) initialize one union-find node for every occurrence environment in $\mathcal{A}$;
(2) construct a spanning forest $G_\varphi$ for the blocks of $\varphi$;
(3) for every edge

$$(F, F') \in G_\varphi$$

and every probe $p \in P$, union the occurrence identifiers

$$T_I(p, F) \qquad \text{and} \qquad T_I(p, F');$$

(4) return the equivalence relation represented by the resulting union-find components.

**Theorem 10.6** (Correctness of the Forest Procedure)**.** *The procedure of Definition 10.5 returns*

$$L_P(\varphi).$$

*Proof.* The selected forest generates $\varphi$, so

$$\mathrm{EqCl}_{\mathcal{E}}\big(G_\varphi\big) = \varphi.$$

The union operations construct

$$\mathrm{EqCl}_{\mathcal{A}}\big(R_{P,G_\varphi}\big).$$

By Theorem 5.6,

$$\mathrm{EqCl}_{\mathcal{A}}\big(R_{P,G_\varphi}\big) = L_P(\varphi).$$

□

Because

$$\Theta_P(\varphi) = L_P(\varphi),$$

the same procedure computes the interpretive threshold.

For a principal pair regime $\gamma_{F,F'}$, the generating forest consists of the single nontrivial edge

$$(F, F').$$

The procedure then computes

$$\delta_P(F, F')$$

using one union operation per probe.

*Remark* 10.7 (Release-Family Reduction). When a frame regime partitions a finite release family, its spanning forest requires one fewer edge than the number of releases in each nonempty block.

The procedure therefore does not compare every release with every other release. It interprets the selected probe family once per release and performs union operations only along the generating forest.

## 10.3 Complexity

Let

$$n_{\mathcal{E}} = |\mathcal{E}|, \qquad n_{\mathcal{A}} = |\mathcal{A}|, \qquad n_P = |P|.$$

For a frame regime $\varphi$, let

$$b_\varphi = |\mathcal{E}/\varphi|$$

be its number of blocks.

Its spanning forest has

$$n_{\mathcal{E}} - b_\varphi$$

edges.

Let

$$C_I$$

denote the total cost of computing and canonicalizing all entries of the interpretation table:

$$C_I = \sum_{F \in \mathcal{E}} \sum_{p \in P} C_I(p, F).$$

This quantity includes domain-specific work such as recurrence generation and chronological sorting.

Let

$$\alpha$$

denote the inverse-Ackermann function arising from union-find operations.

**Theorem 10.8** (Computational Complexity)**.** *Under the finite comparison assumptions:*

*(a) the interpretation table requires*

$$C_I$$

*time and*

$$O\big(n_P n_{\mathcal{E}}\big)$$

*stored occurrence identifiers, in addition to the canonical outputs themselves;*

(b) *after the table and occurrence-regime labels are available, constructing*

$$K_P(\rho)$$

*by hashed signature grouping requires expected time*

$$O\big(n_P n_{\mathcal{E}}\big);$$

(c) *deterministic sorting of the signatures gives the alternative bound*

$$O\big(n_P n_{\mathcal{E}} \log n_{\mathcal{E}}\big);$$

(d) *constructing a spanning forest for* $\varphi$ *requires*

$$O\big(n_{\mathcal{E}}\big)$$

*time when its blocks are already known;*

(e) *computing*

$$L_P(\varphi) = \Theta_P(\varphi)$$

*from that forest requires*

$$n_P\big(n_{\mathcal{E}} - b_\varphi\big)$$

*union operations and therefore*

$$O\left(n_P\big(n_{\mathcal{E}} - b_\varphi\big)\alpha\big(n_{\mathcal{A}}\big)\right)$$

*additional time;*

(f) *computing one pairwise interpretive demand*

$$\delta_P(F, F')$$

*requires*

$$n_P$$

*union operations and therefore*

$$O\left(n_P \alpha\big(n_{\mathcal{A}}\big)\right)$$

*time after interpretation-table construction.*

The algebraic overhead is therefore near-linear in the size of the finite carriers and selected probe family.

The procedure avoids:

(a) factorial enumeration of occurrence matchings;
(b) quadratic enumeration of all pairs within frame-regime blocks; and
(c) repeated interpretation of the same probe under the same frame environment.

The interpretation cost $C_I$ may dominate the partition operations. That is expected. The computational burden lies in producing the domain interpretation, not in the finite lattice algebra used to compare its results.

*Remark* 10.9 (Cost of Canonicalization). Suppose one schedule interpretation result contains $k$ generated occurrences.

Chronological canonicalization may require

$$O(k \log k)$$

sorting time.

Once canonicalized, index-wise processing is linear in $k$.

The rejected matching comparison would instead range over as many as

$$k!$$

bijections, with lattice operations performed for each matching.

*Remark* 10.10 (Incremental Probe Addition). Suppose a kernel has already been computed for a probe family $P$, and a new probe $q$ is added.

The new interpretation values

$$T_I(q, F)$$

are computed once for every $F \in \mathcal{E}$.

The probe-union law gives

$$K_{P \cup \{q\}}(\rho) = K_P(\rho) \wedge K_{\{q\}}(\rho).$$

Threshold construction is likewise incremental:

$$\Theta_{P\cup\{q\}}(\varphi) = \Theta_P(\varphi) \vee \Theta_{\{q\}}(\varphi).$$

The interpretation table may therefore be extended one probe at a time, and the corresponding kernel or threshold updated by one partition meet or join.

## 10.4 Witness Extraction

Every failed admissibility claim has a finite witness.

**Definition 10.11** (Admissibility-Failure Witness)**.** Let

$$\varphi \in \mathrm{Eq}(\mathcal{E})$$

be a frame regime, and let

$$\rho \in \mathrm{Eq}(\mathcal{A})$$

be a proposed occurrence regime.

An *admissibility-failure witness* is a tuple

$$\langle F, F', p, A, A' \rangle$$

such that:

(a)
$$F \;\varphi\; F';$$

(b)
$$A = I(p, F);$$

(c)
$$A' = I(p, F');$$

(d)
$$A \not\rho A'.$$

The witness identifies:

(a) the frame pair the declaration requires to remain equivalent;

(b) the probe exposing the disagreement; and

(c) the two canonical occurrence environments the proposed occurrence regime fails to identify.

**Theorem 10.12** (Finite Admissibility Witness)**.** *The occurrence regime $\rho$ is not admissible for $\varphi$ under $P$ if and only if an admissibility-failure witness exists.*

*Proof.* By Definition 6.1, admissibility fails when

$$\varphi \not\sqsubseteq K_P(\rho).$$

This holds when there exist $F, F' \in \mathcal{E}$ such that

$$F \;\varphi\; F'$$

but

$$F \not\!K_P(\rho)\; F'.$$

By Definition 4.2, the latter condition holds when some $p \in P$ satisfies

$$I(p, F) \not\!\!\not\, I(p, F').$$

These finite objects form the required witness. □

A witness may be extracted without enumerating every pair in $\varphi$.

**Theorem 10.13** (Generating-Forest Witness Extraction)**.** *Let $G_\varphi$ be any spanning forest generating $\varphi$.*

*If $\rho$ is not admissible for $\varphi$, then there exist*

$$(F, F') \in G_\varphi$$

*and*

$$p \in P$$

*such that*

$$I(p, F) \not\!\!\not\, I(p, F').$$

*Proof.* Suppose every forest edge satisfies

$$I(p, F) \ \rho \ I(p, F')$$

for every $p \in P$.

For any two frame environments in the same $\varphi$-block, the forest supplies a finite path between them.

Applying the same probe along that path and using transitivity of $\rho$ shows that their interpreted outputs are $\rho$-equivalent.

Therefore every pair in $\varphi$ lies in $K_P(\rho)$, so

$$\varphi \sqsubseteq K_P(\rho).$$

This contradicts failed admissibility. Hence some forest edge and probe provide the required witness. □

After construction of the interpretation table, the forest-based admissibility test examines at most

$$n_P\big(n_{\mathcal{E}} - b_\varphi\big)$$

output-label pairs.

It stops immediately when the first mismatch is found.

**Definition 10.14** (Kernel-Separation Witness)**.** For frame environments $F, F'$ lying in different blocks of $K_P(\rho)$, a *kernel-separation witness* is a probe $p \in P$ satisfying

$$I(p, F) \not\rho I(p, F').$$

Such a witness is obtained by scanning the two stored signatures until their first unequal $\rho$-block label is found.

**Definition 10.15** (Refinement-Failure Witness)**.** For equivalence relations

$$\alpha, \beta \in \mathrm{Eq}(X),$$

a *refinement-failure witness* for

$$\alpha \not\sqsubseteq \beta$$

is a pair

$$(x, y) \in X \times X$$

such that

$$x \;\alpha\; y \qquad \text{and} \qquad x \not\beta\; y.$$

A refinement witness can be extracted by scanning each $\alpha$-block and checking whether all of its members share one $\beta$-block label.

If two partitions are incomparable, the procedure returns two witnesses:

$$x \;\alpha\; y \qquad \text{and} \qquad x \not\beta\; y,$$

together with

$$u \;\beta\; v \qquad \text{and} \qquad u \not\alpha\; v.$$

For threshold comparison, these pairs state which occurrence equivalence one demand requires and the other omits.

**Theorem 10.16** (Finite Decidability)**.** *Under the finite comparison assumptions, and assuming that the interpretation function and declared canonicalization are computable, the following are computable:*

*(a) the interpretive kernel $K_P(\rho)$;*
*(b) the generated occurrence regime $L_P(\varphi)$;*
*(c) the threshold $\Theta_P(\varphi)$;*
*(d) the pairwise interpretive demand $\delta_P(F, F')$;*
*(e) admissibility of $\rho$ for $\varphi$;*
*(f) refinement, equality, strict refinement, and incomparability of finite regimes; and*
*(g) finite witnesses refuting every failed admissibility or refinement claim.*

The procedure does not discover the correct probe family, frame carrier, occurrence carrier, interpretation signature, canonicalization, or declared regimes.

Those are semantic declarations supplied by the domain realization.

Given those declarations, the procedure computes their consequences and returns finite witnesses when a proposed occurrence regime does not support a declared

frame demand.

# 11 Evidence Plan

The mathematical results of this paper hold for every finite comparison artifact satisfying the stated hypotheses.

The schedule interpretation is intended to establish a different claim: that the construction identifies nontrivial and practically relevant structure in a real domain.

That claim is not established by the algebra alone.

The evidence plan therefore separates:

(a) the proved properties of interpretive kernels;
(b) the prospective classification of a previously assembled schedule-question corpus;
(c) the executable computation of kernels and thresholds over pinned schedule artifacts; and
(d) the residuals that would limit, revise, or disconfirm the proposed schedule realization.

No corpus classification, kernel result, or residual count is reported in this section. Those results must be generated from the frozen evidence artifacts described below.

## 11.1 Question Corpus

The initial evidence base is a corpus of fifty-one schedule-sameness questions assembled before the present classification.

The questions concern recurring rules, exceptions, time-zone changes, holiday and business calendars, copied and overlaid calendars, identifiers, version histories, cancellation and restoration, declared and executed schedules, boundaries, granularity, and frame-dependent interpretation.

The corpus was assembled as a challenge set, not as a random sample from a defined population of schedule questions. It therefore supports finite coverage, separation,

and residual claims about the examined questions. It does not support statistical prevalence claims about scheduling practice as a whole.

**Definition 11.1** (Question-Corpus Entry)**.** A *question-corpus entry* contains:

(a) a stable question identifier;
(b) the question text as originally recorded;
(c) its source or construction provenance;
(d) the date and version at which it entered the corpus;
(e) any examples originally attached to it; and
(f) an amendment history that preserves rather than replaces the original wording.

The original question text is immutable after the classification protocol is frozen. A correction, clarification, or scope restriction is recorded as a new attributed amendment.

The corpus preserves the distinction between observed phenomena and explanatory classifications.

Apparent groupings such as rule versus extension, frame dependence, per-observer copies, identifier versus content, persistence through history, and declared versus executed behavior were recorded as observations.

They are not treated as established regimes, kernel classes, or proof that a new primitive is required.

**Definition 11.2** (Kernel Analysis Record)**.** For each corpus question admitted to kernel analysis, the evidence record declares:

(a) the schedule declaration or declarations;
(b) the finite probe family;
(c) the observation horizon of each probe;
(d) the schedule frame carrier;
(e) the interpretation function and canonicalization procedure;
(f) the candidate frame regime;
(g) the candidate occurrence regime;
(h) the resulting interpretive kernel;
(i) the generated threshold;

(j) the admissibility verdict; and

(k) any finite witness supporting a failed refinement or admissibility claim.

A corpus question need not supply all of these declarations in its original wording.

Missing information is not silently inferred. It is recorded as part of the classification result.

**Definition 11.3** (Question Disposition)**.** Each corpus question receives one primary disposition:

(D1) *kernel-expressible*: the question determines a finite comparison artifact and a declared kernel comparison;

(D2) *object boundary undeclared*: the question does not determine which objects form the comparison carrier;

(D3) *frame regime undeclared*: the question does not determine which frame environments are required to count as equivalent;

(D4) *occurrence regime undeclared*: the question does not determine which generated occurrence environments count as equivalent;

(D5) *probe domain undeclared*: the question omits the horizon, output channel, or other observation declaration required to make interpretation finite;

(D6) *resolution policy presented as regime*: the stated practice selects a frame but does not define an equivalence relation on frames;

(D7) *interpretation incomplete*: the declared inputs do not determine a total interpretation function on the examined artifact;

(D8) *outside kernel scope*: the question concerns another relation, such as causal responsibility, normative authority, execution compliance, or object identity without an interpretive-equivalence claim; or

(D9) *residual*: the question is well scoped but cannot be represented faithfully by the declared kernel construction.

These dispositions are mutually exclusive at the primary level. An evidence record may carry additional secondary diagnoses.

In particular, *object boundary undeclared* is distinct from *frame regime undeclared* and *occurrence regime undeclared*.

The first concerns what the compared objects are. The latter two concern which

equivalence relations are declared over known carriers.

For a kernel-expressible question, the computational result is recorded separately.

**Definition 11.4** (Kernel Result Class)**.** A kernel-expressible question receives one of the following result classes:

(K1) *admissible at threshold*: the examined occurrence regime equals the generated threshold;

(K2) *admissible above threshold*: the examined occurrence regime is strictly coarser than the threshold;

(K3) *inadmissible below threshold*: the examined occurrence regime is strictly finer than the threshold;

(K4) *inadmissible and incomparable*: the examined occurrence regime and threshold are incomparable; or

(K5) *observational collapse*: distinct declared frame environments produce zero pairwise demand under the selected probe family.

The result classes report the relation between a proposed occurrence regime and the generated threshold.

They do not rank the substantive importance of the question or determine which regime an institution ought to adopt.

## 11.2 Predicted Classifications

The evidence record preserves two stages of prediction.

The first stage consists of the competing predictions recorded before the corpus was classified and before the present kernel algebra was established.

Those predictions concern:

(a) how completely the referential regimes of the earlier Structural Explainability work would classify the schedule questions;

(b) whether frame identity would require a new regime, operate as an external context parameter, or reuse the existing regime machinery over frame objects;

(c) whether apparent schedule phenomena would collapse into fewer explanatory distinctions;

(d) whether some established regimes would receive no support from the corpus; and

(e) whether any question would remain well scoped but unclassifiable.

Those predictions remain frozen in their original preregistration artifact and are not rewritten to agree with the results of this paper.

The present work adds a second stage of predictions specific to interpretive kernels.

**Prediction 11.5** (Nondegenerate Schedule Kernels)**.** At least one pinned schedule comparison will produce an interpretive kernel that is neither

$$\bot_{\mathcal{E}}$$

nor

$$\top_{\mathcal{E}}.$$

This prediction tests whether the construction has discriminating power in the schedule domain.

If every examined kernel is discrete, the probes distinguish every frame environment under the selected occurrence regimes.

If every examined kernel is indiscrete, the probes distinguish none.

Either extreme is mathematically valid, but a corpus concentrated at one extreme would weaken the claim that the kernel supplies useful schedule structure.

**Prediction 11.6** (Occurrence-Regime Sensitivity)**.** There exists at least one fixed probe family $P$, frame carrier, and pair of occurrence regimes

$$\rho \sqsubset \sigma$$

for which

$$K_P(\rho) \subsetneq K_P(\sigma).$$

The theorem permits strict dependence but does not guarantee that a selected schedule corpus exhibits it.

The evidence must therefore include at least one concrete frame pair whose interpretive equivalence changes when the occurrence comparison changes.

**Prediction 11.7** (Nontrivial Thresholds)**.** At least one declared schedule frame regime has a threshold that is neither the equality relation nor the indiscrete occurrence regime.

A nontrivial threshold identifies some generated occurrence environments while preserving distinctions among others.

Such a result would show that the schedule demand is not merely "all outputs must be equal" or "all outputs may be collapsed."

**Prediction 11.8** (Incomparable Schedule Demands)**.** The schedule evidence contains two declared frame demands whose generated occurrence thresholds are incomparable under refinement.

The candidate construction separates time-zone-sensitive and calendar-sensitive demands.

The empirical result requires actual pinned schedule declarations, frame artifacts, and generated occurrence environments exhibiting the two incomparable requirements.

**Prediction 11.9** (Policy-Regime Separation)**.** At least one production-style practice initially described as frame sameness will fail equivalence-relation validation and will instead be classified as a resolution policy.

Pin-with-fallback, latest-compatible, and always-latest are candidate practices.

The evidence must test the actual declared relation. The classification is not assigned from the policy's name alone.

**Prediction 11.10** (Frame-Component Interaction)**.** At least one schedule interpretation depends jointly on two components of the frame tuple in a way not recoverable by independently computing and combining per-component kernels.

Business-day interpretation under both a holiday calendar and time-zone rules is a candidate interaction.

This prediction is supported only if a finite witness shows that the complete frame kernel differs from the proposed componentwise construction.

**Prediction 11.11** (Parameter Necessity)**.** The corpus contains questions for which reporting the probe family without the occurrence regime, or the occurrence regime without the probe family, leaves the interpretive-equivalence claim underdetermined.

This is an evidentiary prediction about the questions and declarations, not a restatement of Propositions 4.6 and 4.7.

The finite evidence must show the same schedule material receiving different classifications under the omitted parameter.

The following mathematical laws are not registered as empirical predictions:

(a) monotonicity of $K_P$;
(b) meet preservation by $K_P$;
(c) join preservation by $L_P$;
(d) the adjunction;
(e) probe-union duality; and
(f) existence of finite witnesses under the stated finite hypotheses.

They are proved results.

Executable checks of those laws are conformance tests for the implementation.

A failed executable law indicates an implementation defect, an undeclared change of carrier, or violation of a theorem hypothesis.

## 11.3 Provenance and Preregistration

The evidence package is maintained separately from the mathematical manuscript and the schedule implementation.

The manuscript states the protocol and reports the results.

The schedule implementation computes interpretations, canonical forms, partitions, thresholds, and witnesses.

The evidence repository preserves the frozen inputs, preregistered predictions, execution manifests, and generated result bundle.

**Definition 11.12** (Evidence Release)**.** An *evidence release* contains:

(a) the immutable question corpus;
(b) the original preregistration;
(c) the kernel-specific preregistration;

(d) the schedule declarations;
(e) the probe declarations and finite horizons;
(f) the frame-environment inventory;
(g) the candidate frame regimes;
(h) the candidate occurrence regimes;
(i) the canonicalization declaration;
(j) the pinned interpreter and dependency versions;
(k) the cited external frame artifacts;
(l) the complete interpretation table;
(m) the computed kernels, thresholds, and pairwise demands;
(n) every finite failure witness;
(o) the classification results; and
(p) a manifest identifying the hashes and versions of all included artifacts.

For schedule evidence, the pinned external artifacts include, where applicable:

(a) time-zone database releases;
(b) holiday-calendar releases;
(c) business-calendar releases;
(d) provider or authority artifacts;
(e) schedule vocabulary releases; and
(f) the schedule interpretation implementation.

A human-readable release name is not sufficient provenance. The evidence record includes a stable artifact identifier and content digest.

**Definition 11.13** (Preregistration Boundary)**.** The *preregistration boundary* is the recorded point after which the following items may not be altered within the confirmatory evidence release:

(a) the corpus membership and original question wording;
(b) the prediction statements;
(c) the primary classification dispositions;
(d) the candidate frame and occurrence regimes;
(e) the probe families and horizons;
(f) the canonicalization rules;

(g) the pinned external artifacts; and
(h) the criteria for identifying residuals.

The preregistration is assigned a version, timestamp, and content digest before confirmatory classification begins.

A later correction does not overwrite the frozen record. It creates a new release with an amendment explaining:

(a) what changed;
(b) why it changed;
(c) whether the change occurred before or after inspecting results; and
(d) which conclusions depend on the change.

New questions, new probes, longer horizons, new frame releases, and newly discovered occurrence regimes are permitted after preregistration, but belong to an exploratory extension set.

They do not silently enter the confirmatory corpus.

**Definition 11.14** (Result Provenance Record)**.** Every computed kernel result is accompanied by:

(a) the question identifier;
(b) the probe-family identifier;
(c) the frame-carrier identifier;
(d) the frame-regime identifier;
(e) the occurrence-regime identifier;
(f) the interpretation-table digest;
(g) the implementation version;
(h) the canonicalization version;
(i) the computed partition;
(j) the threshold partition;
(k) the verdict; and
(l) any supporting witness identifiers.

The result record makes explicit the two constitutive kernel parameters.

A statement that reports "the schedule kernel" without identifying both the probe family and occurrence regime is incomplete.

The evidence bundle also preserves negative results.

A question classified as outside kernel scope, a policy rejected as a non-partition, a degenerate kernel, an unexercised regime, or a residual remains in the released result set.

## 11.4 Qualifying Residuals

A residual is not simply a question without enough information.

Missing boundaries, missing regimes, missing horizons, or missing frame artifacts are declaration failures and receive their corresponding dispositions.

A residual remains after the required declarations have been supplied.

**Definition 11.15** (Faithful Residual)**.** A corpus question is a *faithful residual* when:

(a) the compared objects and finite carriers are explicit;
(b) the probe family and finite horizons are explicit;
(c) the frame environments and candidate frame regime are explicit;
(d) the occurrence carrier, canonicalization, and candidate occurrence regimes are explicit;
(e) interpretation is total on the examined artifact;
(f) the question makes a genuine interpretive-equivalence demand;
(g) the demand cannot be represented by the kernel construction without adding, erasing, or misclassifying a distinction present in the question; and
(h) the failure is not repaired by selecting another already-declared frame or occurrence regime.

The faithfulness condition prevents a residual from being eliminated merely by encoding the disputed distinction into a larger carrier or a specially constructed frame variable.

A representation is not faithful when it succeeds only by placing the desired answer into the input.

The following outcomes count as substantive residuals.

(R1) *Missing comparison machinery.* A well-scoped schedule question requires a sameness distinction that no admitted occurrence or frame regime can express.

(R2) *Nonfunctional interpretation.* The complete declared inputs do not determine one interpretation result, and the remaining variation cannot be represented as an explicit frame component or probe parameter.

(R3) *Unfaithful frame reduction.* A frame-sensitive question can be represented only by erasing a distinction present in the source artifacts or adding a distinction not supplied by them.

(R4) *Unrepresentable threshold.* The threshold exists in the ambient lattice $\mathrm{Eq}(\mathcal{A})$ but no occurrence regime admitted by the schedule vocabulary denotes that partition.

(R5) *Invalid component factorization.* A declared componentwise frame model cannot reproduce the kernel of the complete frame tuple, and the interaction cannot be expressed by the existing frame declaration.

(R6) *Persistent scope mismatch.* A question remains interpretive but cannot be placed on one fixed finite comparison artifact without changing the object whose equivalence is being asked about.

The following outcomes do not, by themselves, count as residuals.

(N1) a question whose object boundary is undeclared;

(N2) a question whose occurrence or frame regime is undeclared;

(N3) a question lacking a finite horizon;

(N4) a resolution policy that fails partition validation;

(N5) an inadmissible occurrence regime for which a finite witness is returned;

(N6) two incomparable thresholds;

(N7) a distinct frame pair with zero demand under the selected probes;

(N8) a regime not exercised by the current finite corpus;

(N9) a result outside the scope of interpretive equivalence; or

(N10) an implementation that violates a proved algebraic law.

These outcomes may be important findings, but each has an explanation within the declared construction.

A regime untouched by the corpus bears specifically on domain-relative necessity. It

does not show that the regime collapses or that the broader vocabulary is invalid.

Likewise, two regimes performing the same work on the finite corpus do not establish global equality. They establish only that the selected evidence does not separate them.

The analysis therefore reports three claims independently:

(a) *coverage*: how many corpus questions receive a non-residual disposition;
(b) *necessity within the corpus*: which declared regimes are required by at least one question or computed threshold; and
(c) *non-collapse within the corpus*: which regime pairs are separated by at least one finite witness.

No one of these claims entails the others.

Complete coverage does not establish that every admitted regime is necessary.

A regime exercised by one question does not establish that it is distinct from every other regime.

Failure to separate two regimes on the present corpus does not establish that they are globally equal.

*Remark* 11.16 (Failure Modes of the Evidence Program). The evidence would materially weaken the schedule claim of this paper if:

(a) nearly every computed kernel is discrete or indiscrete;
(b) changing the occurrence regime never changes a kernel or admissibility verdict;
(c) no nontrivial threshold is observed;
(d) every question is resolved only by declaring another parameter, without excluding or constraining any proposed comparison;
(e) the complete frame kernel always reduces trivially to independent component kernels;
(f) the vocabulary cannot express thresholds generated by ordinary schedule questions; or
(g) faithful residuals remain without a common structural diagnosis.

The fourth failure mode is especially important. A framework that answers every question only with "declare another regime" may accommodate every outcome

without explaining any of them.

The evidence must show not only that declarations make questions well posed, but also that the resulting finite construction:

(a) rules out invalid comparisons;
(b) distinguishes admissible from inadmissible regime choices;
(c) produces nontrivial thresholds;
(d) exposes incomparable demands;
(e) rejects policies that are not equivalence relations; and
(f) returns finite witnesses for its negative verdicts.

For the frozen corpus and pinned schedule artifacts, a successful evidence result would establish that the construction supplies a finite, parameter-explicit, nondegenerate, and refutable account of the examined interpretive demands.

# 12 Related Work

This paper draws on three distinct foundations.

- The Structural Explainability series supplies the representational separation between declared structure, interpretive regimes, and operationally induced relations.
- Order theory supplies partition lattices, refinement, Galois connections, and the preservation laws of adjoint maps.
- Generalized-ultrametric and observational-equivalence work supplies close mathematical and semantic precedents for representing families of equivalence relations through pairwise demands indexed by observations.

The schedule realization additionally depends on calendaring, recurrence, and time-zone standards that define the artifacts over which interpretation is performed.

The contribution is not a replacement for any of these foundations. It combines them into a finite construction that transports equivalence requirements between distinct interpreted carriers.

## 12.1 Structural Explainability Foundations

The present work extends the Structural Explainability sequence developed in *Neutral Substrates*, *Referential Regimes*, and *Operational Identity*.

*Neutral Substrates* separates stable referential structure from causal and normative interpretation (Case 2026a). Its foundational layer preserves the referential commitments required for shared use under persistent disagreement, while framework-dependent interpretation is introduced through attributed extensions. That separation supplies the architectural premise of this paper. A schedule declaration and the artifacts to which it refers may remain fixed while different frame environments produce different interpreted occurrence environments. The interpretation is therefore not collapsed into the represented declaration.

*Referential Regimes* makes sameness relative to an explicit identity basis and derives a family of transformation-invariant referential regimes for neutral substrates (Case 2026b). This paper inherits the methodological requirement that a claim of sameness identifies the regime under which it is made. This paper does not import particular referential regimes as occurrence or frame regimes. Occurrence and frame regimes are equivalence relations on the carriers declared by the present comparison artifact.

*Operational Identity* represents declared and implemented rules of sameness as partitions of one finite record domain and compares them in the refinement lattice (Case 2026c). It establishes the use of finite partition comparison, refinement witnesses, and disclosure-relative computation for examining identity-relevant behavior. The present work uses that finite-partition method but does not import the operational-identity audit itself. It does not use implementation-surface registries, sibling-regime classification, regime-substitution verdicts, or disclosure-completeness claims.

Instead, it introduces two distinct carriers: a carrier of frame environments and a carrier of occurrence environments. An interpretation function connects them, and a probe family determines which interpreted outputs participate in the comparison.

The companion *Operational Identity software* provides an executable realization of the earlier finite-audit method (Case 2026d). The computational procedure of this paper is a separate construction for kernels, generated regimes, thresholds, and witnesses.

The broader ontology-engineering literature treats explicit categories, identity conditions, and persistence commitments as central modeling concerns (Arp et al. 2015).

Classification studies likewise emphasize that categories and standards operate as consequential infrastructure rather than as neutral labels (Bowker and Star 1999). The present work is narrower. It does not propose a general ontology of frames or occurrences. It assumes finite typed carriers and studies the equivalence relations induced between them by interpretation.

## 12.2 Partition Lattices and Galois Connections

Equivalence relations on a fixed set form a complete lattice under refinement (Davey and Priestley 2002).

Under the orientation used here, a finer equivalence relation is lower in the order. Meet is intersection, and join is the equivalence closure of union.

This lattice structure supplies the occurrence-regime lattice

$$\mathrm{Eq}(\mathcal{A})$$

and the frame-regime lattice

$$\mathrm{Eq}(\mathcal{E}).$$

Galois connections provide paired monotone maps between ordered structures whose inequalities are equivalent in opposite domains. The general construction was introduced by Ore (Ore 1944) and is now standard in lattice and order theory (Davey and Priestley 2002).

For a left adjoint $L$ and right adjoint $K$,

$$L(x) \sqsubseteq y \quad \Longleftrightarrow \quad x \sqsubseteq K(y).$$

The left adjoint preserves joins, and the right adjoint preserves meets.

Abstract interpretation made lattice-theoretic adjunctions operationally central in program analysis (Cousot and Cousot 1977). There, concrete and abstract semantic domains are related by maps that construct or approximate program properties and fixpoints.

The present adjunction has a different role.

The two domains are lattices of equivalence relations over different carriers.

The right adjoint

$$K_P : \mathrm{Eq}(\mathcal{A}) \longrightarrow \mathrm{Eq}(\mathcal{E})$$

transports an occurrence-equivalence declaration to the frame equivalence that it induces under the probe family.

The left adjoint

$$L_P : \mathrm{Eq}(\mathcal{E}) \longrightarrow \mathrm{Eq}(\mathcal{A})$$

constructs the least occurrence regime required to support a declared frame regime.

The existence and general preservation laws of adjoints are classical. The contribution here is the explicit probe-generated pair

$$L_P \dashv K_P,$$

its finite construction from an interpretation table, and its interpretation as an exact threshold between admissible and inadmissible occurrence regimes.

## 12.3 Lattice-Valued Generalized Ultrametrics

Lawvere recast generalized metric spaces as enriched categorical structures, showing that numerical distance is one instance of a broader ordered notion of comparison (Lawvere 1973).

Prieß-Crampe and Ribenboim developed generalized ultrametric spaces whose distance values lie in partially ordered structures rather than necessarily in the nonnegative real numbers (Priess-Crampe and Ribenboim 1996).

In such spaces, the lattice join plays the role taken by maximum in an ordinary ultrametric inequality. Distances may therefore express qualitatively different and incomparable demands.

A close mathematical precedent for Section 7 is Braunfeld's correspondence between lattice-valued ultrametric spaces and structures equipped with lattice-indexed families of equivalence relations (Braunfeld 2019).

In that correspondence, the distance between two points is constructed from the finest equivalence relation that identifies them, and the equivalence family can be recovered by thresholding those distance values.

The broad correspondence between meet-preserving equivalence families and lattice-valued ultrametrics is therefore prior work. The representation result of Section 7 is a finite interpretive specialization of that correspondence. The specialization has three features specific to the present construction.

First, the value lattice is itself the complete partition lattice

$$\mathrm{Eq}(\mathcal{A})$$

of occurrence environments.

Second, the equivalence family is not supplied independently. It is generated from a finite probe family, an interpretation function, and an occurrence regime.

Third, the pairwise value

$$\delta_P(F, F')$$

has an operational threshold meaning: it is the least occurrence regime under which the selected probes make the two frame environments interpretively equivalent.

Terminology differs across the generalized-metric literature. Some definitions of ultrametric require separation, so that zero distance implies equality.

The present construction does not require that property. Distinct frame environments may have

$$\delta_P(F, F') = \bot_{\mathcal{A}}$$

when the probe family produces identical canonical occurrence environments under both.

Under terminology that reserves *ultrametric* for separating distances, the pairwise interpretive demand is therefore a generalized pseudoultrametric before quotienting by the zero-demand equivalence.

## 12.4 Observational and Testing Equivalence

Testing semantics defines behavioral equivalence through the observations that a selected family of tests can make.

De Nicola and Hennessy show how sets of tests induce distinct process equivalences, including may-testing and must-testing equivalence (De Nicola and Hennessy 1984).

Testing equivalence itself refines the older notion of observational equivalence (Morris 1968; Plotkin 1977), which individuates terms by the contexts that distinguish them.

Two processes are equivalent when the permitted observations cannot distinguish them under the selected testing interpretation. The probe family of this paper has a related observational role.

Two frame environments lie in the same interpretive-kernel block when every selected probe produces occurrence environments that are equivalent under the declared occurrence regime.

There are, however, two substantive differences. First, the observation result is not reduced to a fixed success or failure value. A probe produces a structured occurrence environment, possibly containing ordered values, multiplicity, boundary information, and a terminal tag. Second, equivalence of those results is itself a constitutive parameter. The same probes may induce different frame equivalences under different occurrence regimes.

The construction therefore moves from

equivalence under a family of tests

to

equivalence under a family of probes and a declared output-equivalence regime.

The reverse map has no ordinary analogue in basic testing equivalence. Given a declared frame regime, $L_P$ computes the least occurrence regime under which the probes support that declaration.

The resulting threshold makes explicit which output distinctions must be suppressed to preserve a requested observational equivalence.

Testing equivalence is used only as the nearest semantic precedent for probe-relative indistinguishability; the present setting has no processes or nondeterministic execution, and no process-equivalence machinery is imported.

## 12.5 Canonicalization and Quotient Comparison

Canonical forms are a standard method for making comparison independent of incidental presentation. Term-rewriting theory, for example, uses normal forms to replace repeated comparison over many equivalent representations with comparison of selected representatives (Baader and Nipkow 1998).

The schedule construction uses the same broad strategy.

Raw generated occurrences are placed in a declared chronological canonical form. The occurrence regime is then applied to that indexed representation.

Canonicalization is not treated as a proof that one representation is intrinsically privileged. It is part of the declared comparison protocol and determines the carrier on which the kernel is computed.

Matching and entity-resolution methods address a different problem: finding plausible correspondences among records under imperfect, heterogeneous, or noisy identifying information (Christen 2012).

Such methods may properly optimize over alternative candidate matches.

The rejected matching comparison of Section 8 is not rejected because matching is generally invalid. It is rejected because minimizing over alternative bijections does not preserve the lattice laws required by this construction.

Separate matchings may witness comparison under separate occurrence regimes, while their meet may admit no single matching that witnesses both.

The resulting comparison can fail meet preservation and even fail the strong triangle law.

Permutation invariance alone is therefore insufficient for the present purpose.

The construction requires a declared comparison whose threshold relations are equivalence relations and whose induced kernel preserves meets.

Chronological canonicalization provides that structure for the schedule realization without factorial enumeration of possible occurrence matchings.

## 12.6 Calendar, Recurrence, and Time-Zone Semantics

Internet calendaring standards supply the primary representational foundation for the schedule realization.

RFC 5545 defines the iCalendar data model, recurrence rules, exception dates, floating date-times, UTC date-times, zone-referenced local times, and embedded time-zone components (Desruisseaux 2009).

A recurring component may describe an indefinitely extensible set of occurrences. A referenced time-zone definition must provide valid information for the recurrence instances to which it applies.

These features motivate two explicit declarations in the present model: a finite observation horizon and a frame environment containing the cited time-zone artifact.

JSCalendar provides an independently defined JSON data model for calendar events and tasks and aims to reduce representational ambiguity while remaining broadly compatible with iCalendar semantics (Jenkins and Stepanek 2021).

These standards determine how schedules and calendar values may be represented. They do not determine which distinct representations, artifact releases, or interpreted occurrence sequences must count as equivalent for a particular comparison.

Time-zone interpretation additionally depends on maintained and versioned external artifacts.

RFC 6557 specifies procedures for maintaining the IANA Time Zone Database (Lear and Eggert 2012). RFC 7808 defines a service for distributing time-zone data and associated synchronization information to clients (Douglass and Daboo 2016). RFC 9636 specifies the Time Zone Information Format used to represent and exchange time-zone transition data (Olson et al. 2024).

Together, these standards establish that calendaring systems consume external, mutable, versioned interpretive artifacts.

They do not define an identity regime over releases.

For example, the standards do not decide whether two time-zone releases that have the same current extension but different identifiers or histories are the same frame.

They likewise do not determine:

(a) which frame environments a deployment declaration requires to remain equivalent;
(b) which differences among generated occurrence environments are material;
(c) the least occurrence regime supporting a declared frame regime;
(d) whether a pinning or fallback practice is an equivalence relation rather than a selection policy; or
(e) the finite witness refuting an unsupported frame-equivalence declaration.

Those are the questions addressed by the schedule interpretation of this paper.

Taken together, the related foundations delimit the contribution precisely.

This paper does not introduce partition lattices, Galois connections, adjoint preservation laws, lattice-valued generalized ultrametrics, testing equivalence, canonical forms, recurrence syntax, or time-zone formats.

It contributes:

(1) a finite typed interpretation setting with separate occurrence and frame carriers;
(2) a kernel indexed jointly by a probe family and an occurrence regime;
(3) an explicit adjoint construction producing the least occurrence regime required by a frame regime;
(4) a lattice-valued threshold interpretation of that adjunction;
(5) pairwise interpretive demands as the generalized-pseudoultrametric representation of the resulting kernel family;
(6) finite counterexamples delimiting canonicalization, matching, and partitionhood requirements;
(7) a near-linear partition procedure with finite witness extraction; and
(8) a schedule realization over finite horizons and versioned interpretive frames.

# 13 Limitations

The interpretive-kernel construction is exact relative to a finite declared comparison artifact.

It does not determine which objects, probes, frame environments, occurrence envi-

ronments, canonicalization rules, or equivalence regimes a domain ought to declare.

Those choices remain part of the domain specification. The construction computes their consequences.

## 13.1 Finite Declared Scope

The results assume finite carriers and a finite probe family.

A schedule declaration may denote an indefinitely extensible recurrence, but each probe must supply a finite observation horizon.

The resulting kernel therefore characterizes the selected finite comparison artifact. It does not establish equivalence over all possible probes, all future horizons, or every release of an external frame artifact.

A later probe, longer horizon, or enlarged frame carrier may distinguish environments that the original artifact placed in one kernel block.

## 13.2 Dependence on Probe Family

Interpretive equivalence is observational. Two frame environments may be equivalent under one probe family and distinguishable under another.

Consequently, a reported kernel is incomplete unless it identifies the probe family under which it was computed.

Zero pairwise demand does not establish that two frame environments are identical. It establishes only that the selected probes produce equivalent occurrence environments under the selected occurrence regime.

## 13.3 Dependence on the Occurrence Regime

The occurrence regime is a constitutive parameter. The same interpretation table may induce different frame partitions under different occurrence regimes. The construction does not decide which occurrence distinctions are substantively appropriate. It determines whether a proposed occurrence regime supports a declared frame regime and computes the least regime that would do so.

Admissibility therefore establishes formal compatibility with a declaration, not the correctness of the declaration itself.

## 13.4 Interpretation and Canonicalization

The construction assumes a total interpretation function on the selected finite artifact. It does not resolve semantic ambiguity left undeclared by the domain.

Runtime state, provider behavior, unstated defaults,or incomplete external artifacts must either be represented as declared frame components or excluded from the comparison claim.

Canonicalization is likewise declared. Chronological canonicalization removes incidental generation order, but it may not silently change the represented denotation. A different canonicalization defines a different comparison artifact unless representative independence is proved separately.

The counterexamples of Section 8 show that permutation invariance alone does not justify replacing canonicalization with optimization over alternative matchings.

## 13.5 Frame Completeness

A frame environment may contain multiple interacting artifacts, including time-zone rules, holiday calendars, business calendars, and provider data. The kernel is computed only over the components declared in the frame tuple.

An omitted component can produce apparent equivalence that disappears when the environment is completed.

The present work does not establish that kernels over complete frame tuples factor into independent kernels over individual components. Such factorization requires a separate theorem or finite domain evidence.

## 13.6 Exact Rather than Graded Equivalence

Occurrence and frame regimes are ordinary equivalence relations. The value structure

$$\mathrm{Eq}(\mathcal{A})$$

records exact equivalence requirements, not probabilities, degrees of similarity, confidence values, or fuzzy truth values.

The construction therefore does not directly cover approximate schedule matching, probabilistic interpretation, uncertain external data, or graded equivalence.

An extension to fuzzy equivalences, residuated lattices, or other ordered relation structures would require new definitions and new preservation results.

## 13.7 Nonseparation

The pairwise interpretive demand is generally a pseudoultrametric rather than a separating ultrametric. Distinct frame environments may have

$$\delta_P(F, F') = \bot_{\mathcal{A}}.$$

The construction does not recover representational frame identity from observational equality.

Separation can be obtained only by strengthening the probe family or by quotienting the frame carrier by zero demand.

## 13.8 No Normative Selection Procedure

The kernel does not determine which frame release a system should select, whether an institution should pin or update an external artifact, or which historical, legal, or organizational notion of sameness is preferable. Pinning, latest-compatible selection, always-latest selection, and fallback behavior are resolution policies. The construction can test whether a declared relation is an equivalence relation, but it does not convert a selection policy into an identity regime.

## 13.9 No Implementation-Completeness Claim

The finite procedure of Section 10 operates on the supplied interpretation table and declared carriers. It does not establish that every deployed implementation path uses the declared interpreter, that every external dependency has been disclosed, or that every runtime behavior is represented in the table. Those questions belong to implementation conformance, provenance, and disclosure auditing. A correct kernel computation over incomplete inputs remains correct only for those inputs.

## 13.10 Computational Cost

The partition operations are near-linear in the finite artifact size, but schedule interpretation may be substantially more expensive. Long horizons, dense recurrences, large frame inventories, or many output-rich probes can make construction of the interpretation table the dominant cost. The complexity bounds do not make recurrence expansion constant-time, nor do they prevent the canonical outputs themselves from becoming large.

## 13.11 Evidence Status

The schedule construction of Section 9 shows that the theory admits a nondegenerate finite realization. It is not an empirical result about production schedule systems.

The corpus classification, pinned artifact comparisons, kernel computations, threshold comparisons, and residual analysis described in Section 11 remain separate evidence obligations.

Until those obligations are discharged, the paper establishes the mathematical framework and its executable procedure, not the prevalence or practical distribution of particular schedule kernels.

## 13.12 Domain Transfer

The abstract construction is not restricted to scheduling, but the present domain realization is. Applying it elsewhere requires:

1. finite typed carriers;
2. a total interpretation function on the selected artifact;
3. declared probes;
4. declared equivalence regimes;
5. a comparison-preserving canonicalization; and
6. a faithful account of the external frame environment.

The paper does not establish that every interpretive domain admits these declarations naturally or that the schedule signature generalizes unchanged. A domain in which interpretation is irreducibly probabilistic, nondeterministic, continuous, or dependent on undeclarable runtime state may require a different construction. These limitations delimit rather than negate the principal claim.

Given finite declared carriers, a probe family, a total interpretation function, and an occurrence regime, the interpretive kernel and its least supporting thresholds are computable, their algebraic properties are exact, and failed admissibility claims have finite witnesses.

# 14 Conclusion

An interpreted system contains a rule of sameness for the interpretive frames, a rule of sameness for the outputs produced under those frames, and an interpretation function connecting the two. The interpretive kernel makes that connection a first-class object.

For a finite probe family $P$, the map

$$K_P : \mathrm{Eq}(\mathcal{A}) \longrightarrow \mathrm{Eq}(\mathcal{E})$$

sends an occurrence regime to the frame equivalence that it induces.

Its left adjoint

$$L_P : \mathrm{Eq}(\mathcal{E}) \longrightarrow \mathrm{Eq}(\mathcal{A})$$

sends a frame regime to the least occurrence regime needed to support it.

The adjunction

$$L_P(\varphi) \sqsubseteq \rho \quad \Longleftrightarrow \quad \varphi \sqsubseteq K_P(\rho)$$

turns compatibility between the two declarations into a decidable refinement condition.

It also explains the preservation laws. Interpretive kernels preserve meets because satisfying several occurrence regimes means satisfying their common refinement. Generated occurrence demands preserve joins because supporting combined frame demands requires the equivalence closure of their combined output requirements.

Pairwise interpretive demand provides a second view of the same construction. For two frame environments,

$$\delta_P(F, F')$$

is the least occurrence regime that makes them indistinguishable under the selected probes.

Thresholding these values recovers the kernel family. The values satisfy symmetry, zero self-demand, and the strong triangle inequality in the partition lattice. They need not separate distinct frames, because observational equality under a finite probe family does not imply representational identity.

The finite counterexamples delimit the result. Index-wise comparison does not descend from unordered presentations without canonicalization. Permutation-invariant matching does not guarantee meet preservation or generalized-ultrametric structure. A deterministic frame-selection policy does not become an identity regime unless it is an equivalence relation. These failures identify the conditions under which the kernel construction is well defined and its algebraic guarantees remain valid.

The computational procedure turns the theory into a finite audit. A cached interpretation table supports kernel construction through probe signatures. Generating forests support threshold construction without enumerating every frame pair. Union-find computes the required equivalence closures. Failed admissibility returns a frame pair, a probe, and two occurrence environments that witness the unsupported declaration.

The schedule realization shows how the abstract construction applies when interpretation depends on mutable, versioned external artifacts. A schedule declaration may remain fixed while time-zone, holiday-calendar, business-calendar, or provider artifacts change. Civil output and resolved-instant output remain separate typed dimensions. Finite horizons make recurrence interpretation executable. Chronological canonicalization removes incidental generation order without introducing favorable

matching.

The resulting kernel states which frame environments the selected schedule probes cannot distinguish under the declared occurrence regime. The paper does not determine which regime is substantively correct. It does not choose a time-zone release, holiday authority, resolution policy, or institutional rule of sameness. It computes the consequences of those declarations and identifies the least occurrence demand that each frame declaration imposes. Its guarantees remain relative to the declared carriers, probes, horizons, frame tuple, canonicalization, and occurrence regime.

The evidence plan therefore remains essential. The mathematical construction establishes exact finite laws. The schedule evidence must establish that real comparison artifacts produce nondegenerate kernels, nontrivial thresholds, regime-sensitive classifications, and finite witnesses that matter in practice.

A faithful residual would identify a well-scoped interpretive demand that the declared construction cannot represent without adding or erasing a distinction in the source question. Such residuals would delimit the schedule realization and may motivate a broader semantic structure.

Within its declared finite scope, the result is complete. Given two finite carriers, a finite probe family, a total interpretation function, and an occurrence regime, the induced frame equivalence is computable. Given a frame regime, its least supporting occurrence regime is computable. The two constructions form an adjunction. Their pairwise demands admit a lattice-valued generalized-pseudoultrametric representation. And when a proposed occurrence regime fails to support a declared frame demand, the failure has a finite witness.

Interpretive equivalence is a declared, parameter-relative, and executable relation between rules of sameness on opposite sides of interpretation.

# Statements and Declarations

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### Use of AI-Assisted Tools

AI-assisted tools were used for editing, formatting, and consistency checking. The author reviewed all suggestions and is solely responsible for the content.

### Declaration of Conflicting Interest

The author declares no competing interests.

## References


R. Arp, B. Smith, and A. D. Spear. *Building Ontologies with Basic Formal Ontology*. MIT Press, 2015.

F. Baader and T. Nipkow. *Term Rewriting and All That*. Cambridge University Press, Cambridge, UK, 1998.

G. C. Bowker and S. L. Star. *Sorting Things Out: Classification and Its Consequences*. MIT Press, Cambridge, MA, 1999.

S. Braunfeld. Λ-ultrametric spaces and lattices of equivalence relations. *Algebra Universalis*, 80(3), 2019. doi:10.1007/s00012-019-0606-4.

D. M. Case. *Neutral Substrates: A Design Constraint for Shared Records Under Persistent Interpretive Disagreement*. arXiv preprint arXiv:2601.14271, 2026.

D. M. Case. *Referential Regimes: Transformation-Based Identity Profiles for Neutral Substrates*. arXiv preprint arXiv:2601.16152, 2026.

D. M. Case. *Operational Identity: A Finite Audit of Declared and Implemented Rules of Sameness*. arXiv preprint arXiv:2607.20729, 2026.

D. M. Case. *Structural Explainability Verification: Operational Identity*. Software, version 0.1.0, 2026. doi:10.5281/zenodo.21499599.

P. Christen. *Data Matching: Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection*. Springer, Berlin, 2012.

P. Cousot and R. Cousot. Abstract interpretation: A unified lattice model for static analysis of programs by construction or approximation of fixpoints. In *Proceedings of the 4th ACM SIGACT-SIGPLAN Symposium on Principles of Programming Languages*, pages 238–252, 1977. doi:10.1145/512950.512973.

B. A. Davey and H. A. Priestley. *Introduction to Lattices and Order*. Second edition. Cambridge University Press, Cambridge, UK, 2002.

R. De Nicola and M. C. B. Hennessy. Testing equivalences for processes. *Theoretical Computer Science*, 34(1–2):83–133, 1984. doi:10.1016/0304-3975(84)90113-0.

B. Desruisseaux, editor. *Internet Calendaring and Scheduling Core Object Specification (iCalendar)*. RFC 5545, Internet Engineering Task Force, September 2009. doi:10.17487/RFC5545.

M. Douglass and C. Daboo. *Time Zone Data Distribution Service*. RFC 7808, Internet Engineering Task Force, March 2016. doi:10.17487/RFC7808.

N. Jenkins and R. Stepanek. *JSCalendar: A JSON Representation of Calendar Data*. RFC 8984, Internet Engineering Task Force, July 2021. doi:10.17487/RFC8984.

F. W. Lawvere. Metric spaces, generalized logic, and closed categories. *Rendiconti del Seminario Matematico e Fisico di Milano*, 43:135–166, 1973. doi:10.1007/BF02924844.

E. Lear and P. Eggert. *Procedures for Maintaining the Time Zone Database*. BCP 175, RFC 6557, Internet Engineering Task Force, February 2012. doi:10.17487/RFC6557.

J. H. Morris. *Lambda-Calculus Models of Programming Languages*. PhD thesis, Massachusetts Institute of Technology, 1968. https://hdl.handle.net/1721.1/149376.

A. D. Olson, P. Eggert, and K. Murchison. *The Time Zone Information Format (TZif)*. RFC 9636, Internet Engineering Task Force, October 2024. doi:10.17487/RFC9636.

Ø. Ore. Galois connexions. *Transactions of the American Mathematical Society*, 55:493–513, 1944. doi:10.1090/S0002-9947-1944-0010555-7.

G. D. Plotkin. LCF considered as a programming language. *Theoretical Computer Science*, 5(3):223–255, 1977. doi:10.1016/0304-3975(77)90044-5.

S. Prieß-Crampe and P. Ribenboim. Generalized ultrametric spaces I. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 66:55–73, 1996.

David A. Schmidt. *Denotational Semantics: A Methodology for Language Development*. Allyn and Bacon, Boston, MA, 1986.